Federated Learning-Empowered Mobile Network Management for 5G and Beyond Networks: From Access to Core

Joohyung Lee, *Senior Member, IEEE*, Faranaksadat Solat, Tae Yeon Kim, and H. Vincent Poor, *Life Fellow, IEEE*

***Abstract*—The fifth generation (5G) and beyond wireless networks are envisioned to provide an integrated communication and computing platform that will enable multipurpose and intelligent networks driven by a growing demand for both traditional end users and industry verticals. This evolution will be realized by innovations in both core and access capabilities, mainly from virtualization technologies and ultra-dense networks, e.g., software-defined networking (SDN), network slicing, network function virtualization (NFV), multi-access edge computing (MEC), terahertz (THz) communications, etc. However, those technologies require increased complexity of resource management and large configurations of network slices. In this new milieu, with the help of artificial intelligence (AI), network operators will strive to enable AI-empowered network management by automating radio and computing resource management and orchestration processes in a data-driven manner. In this regard, most of the previous AI-empowered network management approaches adopt a traditional centralized training paradigm where diverse training data generated at network functions over distributed base stations associated with MEC servers are transferred to a central training server. On the other hand, to exploit distributed and parallel processing capabilities of distributed network entities in a fast and secure manner, federated learning (FL) has emerged as a distributed AI approach that can enable many AI-empowered network management approaches by allowing for AI training at distributed network entities without the need for data transmission to a centralized server. This article comprehensively surveys the field of FL-empowered mobile network management for 5G and beyond networks from access to the core. Specifically, we begin with an introduction to the state-of-the-art of FL by exploring and analyzing recent advances in FL in general. Then, we provide an extensive survey of AI-empowered network management, including background on 5G network functions, mobile traffic prediction, and core/access network management regarding standardization and research activities. We then present an extensive survey of FL-empowered network management by highlighting how FL is adopted in AI-empowered network management. Important lessons learned from this review of AI and FL-empowered network management are also provided. Finally, we complement this survey by discussing open issues and possible directions for future research in this important emerging area.**

***Index Terms*—Federated learning, 5G, 6G, network management, artificial intelligence, machine learning.**

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Joohyung Lee and Faranaksadat Solat are with the Department of Computing, Gachon University, Seongnam 13120, Republic of Korea (e-mail: j17.lee@gachon.ac.kr; faranak1995@gachon.ac.kr).

Tae Yeon Kim is with the Network Intelligence Research Section, Electronics and Telecommunications Research Institute, Daejeon 34129, Republic of Korea (e-mail: tykim@etri.re.kr).

H. Vincent Poor is with the Department of Electrical and Computer Engineering, Princeton University, Princeton, NJ 08544 USA (e-mail: poor@princeton.edu).

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I. Introduction

THE FIFTH generation (5G) and beyond wireless networks can be realized by innovations in both core (i.e., virtualization technologies) and access capabilities (i.e., ultra-dense networks) aiming for multipurpose and intelligent networks on an integrated communication and computing platform. Specifically, mature virtualization technologies in the mobile core network, such as software-defined networking (SDN), network slicing, and network function virtualization (NFV), have been widely adopted on top of distributed multi-access edge computing (MEC) for providing the flexibility necessary for dynamic resource allocation [13]. In the access network, an ultra-dense network can be considered an emerging technology for serving a massive number of users with considerable capacity, which also can be easily integrated with extremely high frequencies such as millimeter wave (mmWave) and terahertz (THz) communications [14]. However, those technologies require increased complexity of resource management (e.g., resource allocation, interference management, network deployment, backhauling, congestion management, etc.) [14], and large configurations of network slices [13].

Recently, with the renaissance and democratization of artificial intelligence (AI) and machine learning (ML) algorithms, AI-empowered mobile network management has been attracting considerable attention from industry and academia. Along with this trend, the 3rd generation partnership project (3GPP), an organization that standardizes mobile communication systems, has introduced the network data analytics function (NWDAF) at its technical specifications as a new type of standard control plane (CP) function within the core networks system. The newly proposed NWDAF gathers various data generated from multiple distributed network functions

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(NFs) and provides data analytics (i.e., both training and inference) using ML/AI algorithms [15], [16]. However, due to the increasing number of NFs over distributed base stations (BSs) associated with MEC servers, traditional centralized ML/AI methods, which collect raw training data samples at a centralized core NF, face two challenging problems. That is, 1) consuming enormous network resources and 2) causing data security problems. Hence, for these reasons, with consideration of the distributed nature of NFs, it is necessary to adopt distributed learning algorithms that enable privacy-enhanced and cost-effective AI-empowered mobile network management systems by distributed NFs cooperatively building the shared model. Moreover, such a shared model can anticipate future states (e.g., traffic load, computational load, etc.) and achieve automatic network management.

The concept of federated learning (FL), one of the most promising distributed AI approaches, has been proposed for designing AI-empowered mobile network management in a cost-effective and privacy-enhanced manner [17], [18], [19], [20], [21]. In this sense, under the distributed nature of FL, multiple distributed NFs over distributed BSs associated with MEC servers train their local models by using their local data. Then, the global AI model is constructed by the repeated aggregation process of locally trained models. For instance, leveraging the augmented data collected from distributed BSs associated with MEC servers, a quasi-global mobile traffic prediction model can be constructed via the FL process as in [22]. Specifically, in this work, each MEC server trains its local model by utilizing its local mobile traffic pattern dataset and transmits the computed local model parameters (e.g., weights or gradients) to a centralized server. Then, the centralized server aggregates all local model updates and constructs an improved global model for mobile traffic prediction. Compared to conventional centralized training approaches, adopting FL for AI-empowered mobile network management over distributed NFs has the following advantages.

- Network efficiency: Instead of transmitting raw data to the centralized server, less information, such as model parameters, is delivered to the centralized server. Thus, the data transmission burden for generating the model over the distributed NFs is significantly reduced.
- Privacy: In distributed NFs, to support multi-tenant network management, privacy-sensitive data can be generated over distributed NFs. In FL, since the raw data is kept in local data storage, the leakage of privacy-sensitive data to unintended operators can be minimized. Thus, with such privacy protection features, the AI model for network management can be efficiently improved by utilizing distributed and plentiful datasets over multiple NFs, which might be associated with multiple operators.
- Low latency: In addition to less amount of data transmission, inference can be made at distributed NFs by utilizing the global model. Thus, low transmission delay and real-time decision-making for network management can be achieved by adopting the FL framework.

By leveraging the distributed datasets and computation power of multiple MEC servers with reduced privacy concerns, FL has been applied for use in a variety of AI-empowered mobile network management problems, such as mobile traffic prediction, core, access network management, etc. For instance, FL has been applied to mobile traffic prediction by creating prediction models without sharing traffic load data across multiple BSs associated with MEC servers [22]. The emerging interest in integrating FL and AI-empowered mobile network management reveals that it is the right moment to attract more attention to this promising research area toward 5G and beyond networks.

A. Comparison and Our Contributions

Driven by the recent advances in FL for AI democratization, several survey papers have appeared. In the beginning, several articles about FL in [18], [23], [24], [25], [26], [27] presented the concept of FL with a basic introduction to definitions, architectures, software, platforms, and protocols. Moreover, they provided a few possible research challenges in FL deployment. Recently, the potential of FL in mobile networks with the help of edge computing systems has given rise to various research challenges. Specifically, to design efficient FL in mobile edge networks, a survey in [6] discussed underlying communication costs for executing FL framework and presented various approaches to resource management in FL under mobile edge networks such as participant selection, joint radio and computation management, adaptive aggregation, and incentive mechanisms. In addition to FL on mobile edge networks, the authors also discussed some research challenges in FL for mobile edge networks that suggest that FL can be applicable to help the operation of mobile edge networks. The study in [1] analyzed the use of FL for 5G communications, including edge computing and caching management, spectrum management, and 5G core management via NWDAF. Other studies of [2], [3], [4], [5], [9] investigated the use of FL for 6G applications. Here, the key focuses are on the discussion of designing efficient FL management to support 6G applications in 6G networks or access network management by adopting FL for generating prediction models (e.g., resource management, channel state information (CSI), etc.). The potential of FL in vehicular networks, Unmanned Aerial Vehicle (UAV) networks, and Internet of Things (IoT) networks were explored in [10], [11], [12] by considering applications of different target IoT networks and specific research directions. The integration of FL and SDNs was explored in [8], where the key challenges of using FL in SDNs, such as incentive mechanisms, privacy, security, and model aggregation, are reviewed. Nevertheless, most of these surveys focus on how FL can help applications in different types of mobile networks in an efficient manner. Correspondingly, efficient FL mechanisms are suggested to adopt FL for supporting various applications while guaranteeing the required quality-of-service (QoS). To the best of our knowledge, there is no comprehensive survey that deals with FL-empowered mobile network management, which motivates this work. We present a list of common abbreviations for reference in Table I. Moreover, a comparison of the related work and this paper is given in Table II.

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TABLE I
LIST OF COMMON ABBREVIATIONS

Abbreviation	Description
AF	Application functions
AI	Artificial intelligence
AMF	Access and Mobility Management Function
AnLF	Analytics logical function
ASIC	Application-Specific Integrated Circuits
BS	Base station
CFL	Centralized federated learning
CNN	Convolutional neural network
CQI	Channel quality indicator
CSI	Channel state information
DFL	Decentralized federated learning
DQN	Deep Q-networks
DRL	Deep reinforcement learning
DoS	Denial-of-Service
eMBB	Mobile Broadband
FL	Federated Learning
FPGA	Field-programmable Gate Array
FRL	Federated reinforcement learning
FTL	Federated transfer learning
GAN	Generative adversarial network
gNB	Next-generation NodeB
GPU	Graphics Processing Units
HFL	Hierarchical federated learning
HFCL	Hybrid federated and centralized learning
IID	Independent and Identically Distributed
IoE	Internet of Everything
IoT	Internet of Things
KPI	Key Performance Indicator
LoRa	Long Range
LTE	Long-term evolution
LSTM	Long short-term memory
MDAF	Management data analytics function
MF	Management function
MDP	Markov decision process
MEC	Multi-access edge computing
MIMO	Multiple-input multiple-output
mmWave	Millimeter wave
MTLF	Model training logical function
mMTC	Massive Machine Type Communications
MVNO	Mobile virtual network operators
NRF	NF repository function
NF	Network function
NFV	Network function virtualization
NSSF	Network slice selection function
NWDAF	Network data analytics function
non-RT RIC	Non-realtime RAN intelligent controller
near-RT RIC	Near real-time RAN intelligent controller
OAM	Operations, administration, and maintenance
OPEX	Operating expenses
OTT	Over-the-top
P2P	Peer-to-peer
PCF	Policy control function
PDU	Protocol Data Unit
QoE	Quality of Experience
QoS	Quality-of-service
RB	Resource block
RL	Reinforcement learning
RNTI	Radio network temporary identifiers
RAN	Radio access network
RSSI	Received signal strength indicator
RSRP	Received signal reference power
SaaS	Software-as-a-Service
SDN	Software-defined networking
SNNs	Spiking neural networks
SMF	Session Management Function
SON	Self-organizing networks
SVM	Support vector machine
SLA	Service level agreement
SFL	Split federated learning
THz	Terahertz
UE	User equipment
UPF	User Plane Function
URLLC	Ultra Reliable Low Latency Communications
VFL	Vertical federated learning
VM	Virtual machine
XAI	Explainable AI

B. Structure of the Survey

This survey is organized as shown in Fig. 1. Section II discusses the state-of-the-art of FL, where the fundamental concepts of FL, hardware architecture, networking requirements, FL classification, and FL management are investigated. In Section III, the state-of-the-art of AI-empowered mobile network management is reviewed. Here, we provide an extensive background on AI-empowered mobile network management, including recent 3GPP standard specification status (i.e., NWDAF), networking load/status prediction, networking and computing resource management perspectives, respectively. Then, we discuss recent studies of the integration of FL and AI-empowered mobile network management, which we call FL-empowered mobile network management, in Section IV. From the comprehensive survey, we summarize and highlight several key lessons learned in Section V. The key research challenges and directions are introduced in Section VI. Finally, we conclude the survey in Section VII.

II. FL: STATE OF THE ART

In this section, we review the state-of-the-art of FL as background for this survey. The key concept of FL and its classifications are first discussed. Then, research on FL management for deploying FL in an effective manner is reviewed.

A. Key Concept of FL

The proliferation of primary computing devices such as smartphones and tablets has led to the generation and accumulation of an unprecedented amount of data on these devices. Federated learning (FL), a term coined in 2016 by MacMahan et al., obviates the need to store data on the centralized server and has a distributed training paradigm that allows users to collectively construct a global model by aggregating their locally trained models [17]. By offering new AI solutions in a distributed and privacy-preserving manner, FL has the potential to elevate AI-empowered mobile network management toward 5G networks and beyond. This is because the current 5G technology already includes diverse virtualized NFs distributed on multiple BSs or MEC servers associated with those BSs in a container. This means that the enormous amounts of data over these distributed NFs are collected and need to be analyzed for efficient mobile network management. Correspondingly, the success of FL with respect to mobile device management for diverse mobile applications could naturally be transferred and adapted to build AI-empowered mobile network management in a distributed and secure manner. This could be accomplished by pushing training functions, e.g., NWDAF, to the distributed NFs, where data are generated, rather than to the centralized server.

The typical architecture and training process of an FL involving N participants is depicted in Fig. 2, where the FL server is the model owner and the FL participants are contributors. In this sense, the FL server, a centralized server, coordinates FL tasks over FL participants. Then, each FL participant conducts local training on its local dataset and transmits the local update to the FL server. In the end, the FL server aggregates local updates from the FL participants and generates the global model. This overall procedure can be categorized into the following three steps:

- Step 1: In step 1, the FL server conducts task initialization by providing an initial shared global model w_G^0 and optional information (e.g., incentive info) for local

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TABLE II
EXITING SURVEYS ON FL FOR VARIOUS TYPES OF NETWORKS RELATED TOPICS AND OUR NEW CONTRIBUTIONS

Related Works	Topic	Key Contributions	Limitations
[1]	FL and 5G communication	A brief overview on the use of FL for 5G communications. Specifically, edge computing and caching management, spectrum management, and 5G core management via NWDAF.	The paper only focuses on providing a brief overview of the use of FL for 5G communications. An extensive survey for FL-empowered mobile network management has not been presented.
[2]	FL and 6G communication	A brief survey about research challenges and future trends of efficient FL for 6G communications.	The paper provides a brief overview of the resource management and security issues when applying FL to wireless networks. An extensive survey for FL-empowered mobile network management has not been presented.
[3]	FL and 6G communication	A survey on the use of FL for 6G applications. They also discussed efficient FL management in 6G networks when applying FL to wireless networks.	The use of FL-empowered mobile network management was not considered.
[4]	FL and 6G communication	A brief survey on the FL-base station architecture for 6G applications.	Only FL-base station architecture for 6G application is briefly studied by considering mobile devices as FL participants. The use of FL-empowered mobile network management is not covered.
[5]	FL and 6G communication	Overview of studies on FL in wireless networks and a brief survey on driving FL applications for 6G wireless problems	FL applications for 6G wireless problems are confined to the access network issues. Core network management by adopting FL was not surveyed.
[6]	FL and mobile edge networks	A survey on approaches to resource management in FL under mobile edge networks such as participant selection, joint radio and computation management, adaptive aggregation, and incentive mechanisms. Privacy and security issues in FL are described. FL-base station approaches for mobile edge network optimization are discussed and might slightly overlap with our survey.	The paper mostly focuses on resource management and privacy issues in FL. Discussion on FL-base station network management is confined to mobile edge network optimization, whereas FL-empowered mobile network management is not comprehensively surveyed.
[7]	FL and edge computing	A systematic survey of the use of FL in edge computing systems, including resource management of FL. Specifically, they introduced various applications of FL in edge computing and its implementation issues.	The paper mostly focuses on resource management and application for FL in the edge computing environment. A comprehensive survey on FL-empowered mobile network management was not conducted.
[8]	FL and SDNs	A survey on the use of FL in SDNs. Specifically, related mechanisms and solutions, including incentive mechanisms, privacy, security, and model aggregation, which enable FL in SDNs in an effective manner are reviewed.	The paper only focuses on the adoption of FL in SDNs.
[9]	FL for physical layer design	A brief survey on the recent advances in FL-based training for physical layer design problems.	The paper only focuses on physical layer design problems when FL-based training is considered.
[10]	FL and IoT	A survey on the integration of FL-IoT, with a discussion of the roles of FL in various IoT services and IoT applications.	The paper only focuses on discussing the roles of FL for IoT services and applications by providing insights into FL-IoT integration.
[11]	FL and IoT	A brief survey on the use of FL in IoT. They also discussed the technical challenges of applying FL to vehicular IoT.	The paper only focuses on discussing the roles of FL in vehicular IoT.
[12]	FL and UAVs	A brief survey on the use of FL in UAV networks.	The paper only focuses on the adoption of FL in UAV networks
Our paper	FL-empowered mobile network management for 5G networks and beyond	An extensive survey on FL-empowered mobile network management. Particularly, • We begin with a brief tutorial on FL from multiple viewpoints, offering insights into its potential benefits in mobile network management. Next, we delve into the role of AI as a foundational technology in this domain, representing recent 3GPP standardization efforts as practical perspectives. Our discussion then shifts to an in-depth review of existing approaches for addressing diverse issues in mobile network management explored in the literature. The pros and cons of these approaches are investigated, aiming to pave the way for integrating FL in this field. • We extensively discuss the roles of FL in various AI-enabled network management tasks for 6G, i.e., mobile traffic prediction, core/access network management such as computing and networking provisioning for NFs, network slice configuration, admission control for network slice request, etc. As a practical perspective, we also discuss the role of FL in standards specification. • The tables offer a comprehensive summary of research activities, encompassing a variety of issues and their associated approaches. The FL methods and FL management techniques used in these approaches, along with a thorough analysis of their pros and cons are investigated. Based on this analysis, key lessons learned are outlined, offering insights into FL-empowered mobile network management for 5G and beyond networks. Moreover, research challenges and directions are suggested.	-

training to multiple FL participants. The initial shared global model might include a TensorFlow graph, weights, and instructions. In this step, the FL server might select an appropriate set of FL participants rather than requesting the entire pool of participants to contribute to the FL process. Random selection or various selection algorithms can be proposed for efficient FL. Moreover, incentive mechanisms by the FL server (i.e., FL service provider) can be introduced to promote the participation of the FL participants in the FL process. Then, relevant information

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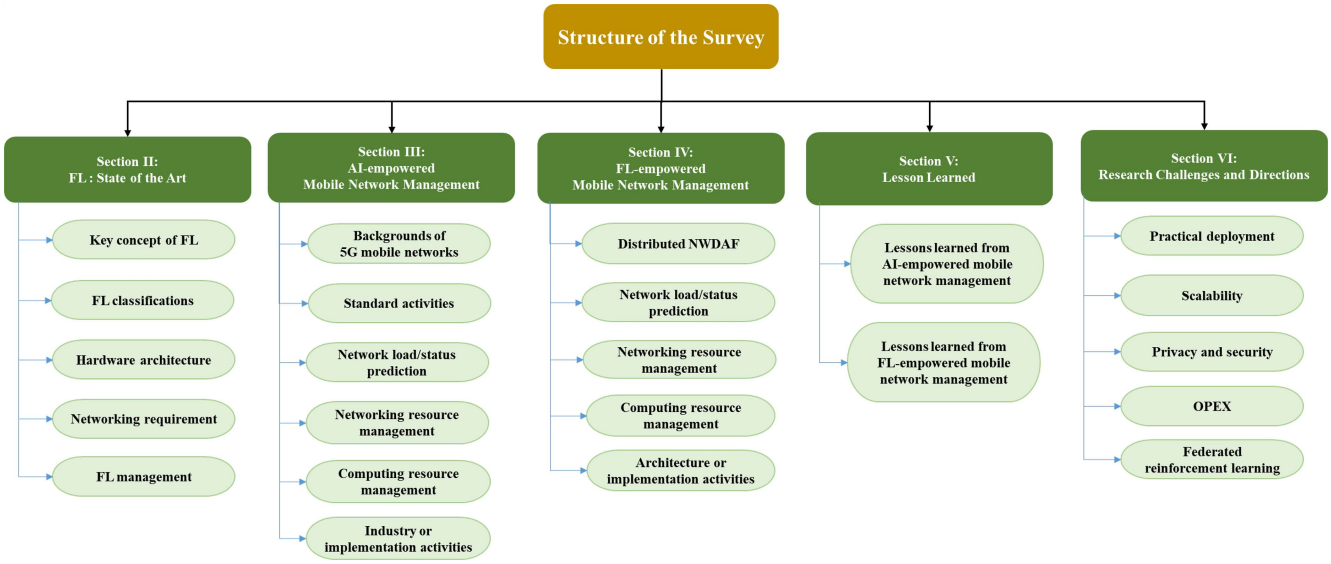


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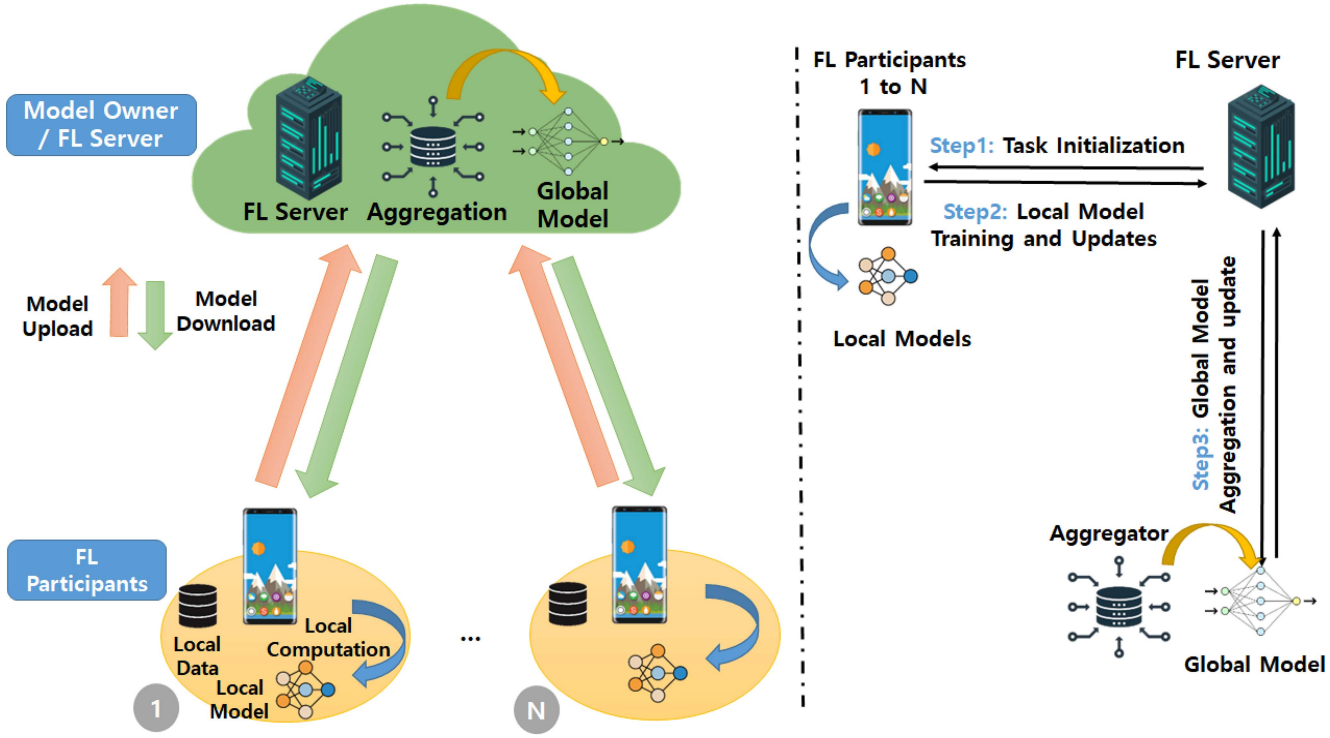


Fig. 2. General FL training process involving N participants [6].

on incentives and incentive mechanisms can also be proposed.

- Step 2: Each FL participant i conducts local training on the entire local data D_i using the shared global model (i.e., w_G^0 in the initial round or w_G^t in round t) where I is the set of FL participants. Specifically, by minimizing the loss function $\mathcal{L}(w_i)$, the local model parameter w_i at FL participant i is given by

$$w_i^* = \arg\min_{w_i} \mathcal{L}(w_i). \tag{1}$$

Then, the updates are transferred to the FL server. Optionally, rather than using the entire dataset, the FL participants might determine the portion of the dataset they use for extracting sample data from the entire dataset based on their interests. In this step, to achieve both communication- and computation-efficient FL, both the i) radio resources for communication and ii) computation resources for local training over FL participants can be adjusted. Many studies have investigated these aspects. Generally, although most FL studies focused on mobile devices as FL participants, in this survey,

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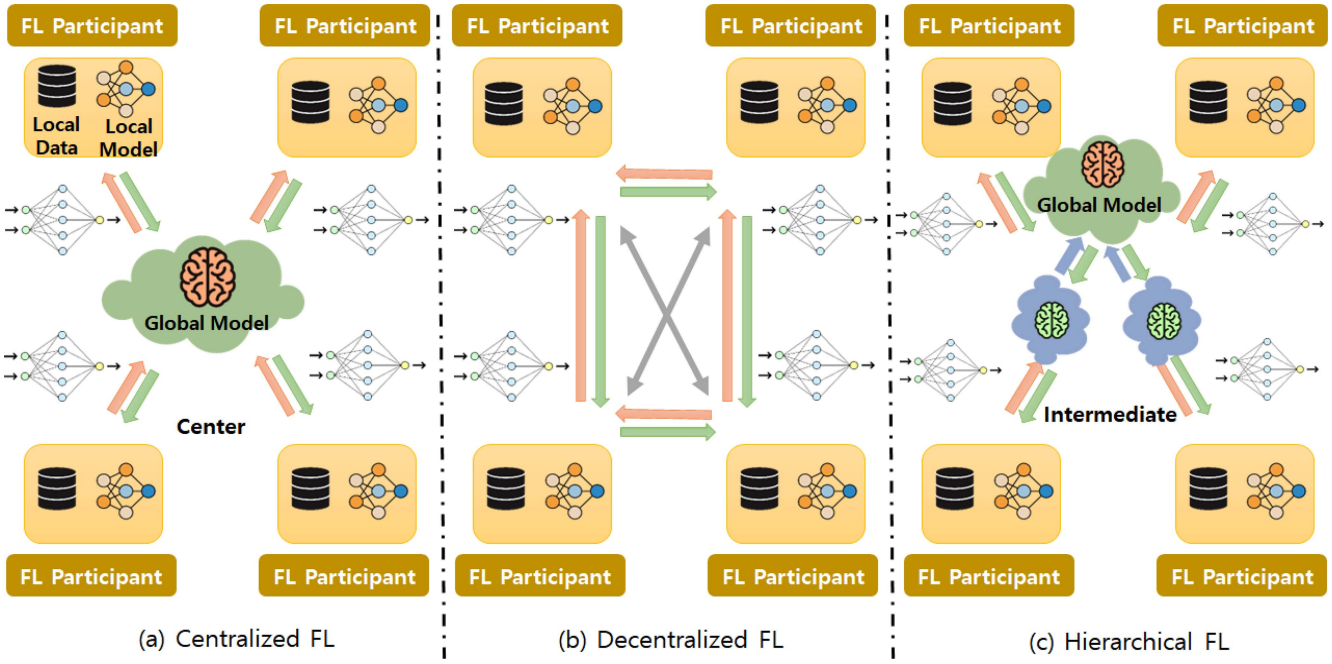


Fig. 3. Networking structures of FL.

distributed NFs, BSs, or MEC servers become FL participants to gather relevant network-level training datasets and conduct network management at a different level (e.g., access network management, core network management). Specifically, in the case of mobile devices, the underlying assumption that the data owners are honest may not always be practical. Thus, many researchers have attempted to solve this problem. On the other hand, such an assumption can always be held in the case of NFs, BSs, or MEC servers managed by trustful service providers.

- Step 3: The FL server merges the local model updates from the FL participants and generates a global model by minimizing the global loss function $\mathcal{L}(w_G^t)$, i.e.,

$$\mathcal{L}(w_G^t) = \frac{1}{N} \sum_{i \in I} \mathcal{L}(w_i). \tag{2}$$

Then, the FL server sends the updated global model parameters back to the FL participants.

Multiple rounds of the FL process, including Steps 2-3, are repeated until the global loss function at the FL server converges to the termination condition, or until the specified target accuracy is reached. Note that in the FL process, different types of ML models can be chosen depending on the application of the FL service. At the same time, many studies on FL utilized the convolutional neural network (CNN) model to obtain an image classification model. Various types of neural networks (CNN, long short-term memory (LSTM) model, etc.), regression models (e.g., linear regression), K-means clustering models, support vector machine (SVM) models, etc. can be used for the FL process. Furthermore, in Step 3, the global model generation, which is an integral part of FL, could be achieved by applying various algorithms for different purposes including safeguarding user privacy, enhancing convergence rates, and mitigating the harm caused by malicious agents [6], [28]. For instance, the simplest algorithm is Average Aggregation, which calculates the mean of updates from all FL participants. However, it may not perform optimally when dealing with non-Independent and Identically Distributed (IID) data. To improve upon this, Weighted Aggregation, such as the Federated Averaging algorithm (FedAvg), assigns varying weights to contributions from different FL participants based on performance or other factors (e.g., dataset size). Secure Aggregation employs techniques such as homomorphic encryption or secure multi-party computation to provide a high level of privacy protection. Nevertheless, it requires the cost of significant computational overhead, leading to slower convergence compared to other aggregation algorithms. Another approach to this perspective, the Differential Privacy Average Aggregation algorithm, ensures strong privacy protection by injecting random noise into the model updates before aggregation. However, it may still exhibit reduced speed and efficiency compared to other aggregation methods. There are numerous other aggregation algorithms in the literature, each with its own set of advantages and drawbacks. Some may be better suited for specific FL contexts than others. Hence, a thoughtful selection of aggregation algorithms is essential, considering the specific purpose of the services [28].

B. FL Classifications

Based on recent studies, we classify FL into three dimensions, namely, networking structure, synchronization approach, and data partitioning [10], [29].

1) Networking Structure: The networking structures for FL can be categorized into three classes, including centralized FL, decentralized FL, and hierarchical FL [10], [29], [30], [31] as summarized in Fig. 3.

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- Centralized FL (CFL): As exemplified in Fig. 3, the CFL is one of the most popular networking architectures for FL because it is simple and easy to deploy in practice. The initial study aiming to implement FL is based on this CFL [19]. In CFL, similar to the client-server architecture, a single CFL server is involved in the aggregation process to construct a global model from the local models generated by FL participants. Even though this architecture has many advantages in terms of deploying FL with a less complex management protocol, these structures often experience long local and global model transmission delays depending on the physical distance and network status between the centralized server and FL participants. Moreover, the bandwidth at the CFL servers might become overloaded, which degrades the convergence speed. Obviously, the risk of a single point of failure in this centralized architecture always exists.
- Decentralized FL (DFL): The DFL was proposed to address the above-mentioned problem. In this structure, the model parameters of FL participants are exchanged only among interconnected FL participants without using a centralized server. Similar to peer-to-peer (P2P) architecture, all FL participants are connected in a P2P manner to generate the global model, which requires more complex management to achieve faster model convergence than CFL. Academic efforts have widely relied on the blockchain protocol to implement DFL in a practical manner.
- Hierarchical FL (HFL): The HFL was proposed to take advantage of the merits of both CFL and DFL. The topology for HFL can be managed in a dynamic manner [29], or a static manner [32]. In HFL, localized aggregations at the intermediate aggregation node can avoid high communication costs and alleviate the burden at the centralized server. Furthermore, to boost the convergence speeds, clustering FL participants for intermediate aggregation in this structure can be widely studied to prevent model deviation under non-IID data [22].

2) Synchronization Approach: The synchronization approach for FL can be categorized into three classes, including strong synchronization, asynchronization, and weak synchronization [29].

- Strong synchronization FL: In strong synchronization FL, the model aggregation frequencies of all FL participants are the same. However, if participants have different computing power or different amounts of data, local training speeds may vary. For example, faster participants would have to wait for slower participants until their models are aggregated in a synchronized manner.
- Asynchronous FL: To address this issue, asynchronous FL allows participants to independently push updates to the central FL server. This enables the global model to be updated whenever new, updated local models are received, without the need to synchronize all local models from all FL participants. This approach can eliminate stragglers and efficiently utilize computing resources, but it may degrade the convergence speed due to the severe staleness problem when data is non-IID and unbalanced.
- Weak synchronization FL: In weak synchronization, FL participants update their local models at different frequencies before model aggregation on the aggregation server in a synchronized manner. This architecture has recently been well-investigated in HFL architecture by addressing the trade-off between strong synchronization FL and asynchronous FL.

3) Data Partitioning: Data partitioning for FL can be categorized into three classes, including horizontal FL, vertical FL, and federated transfer learning (FTL) [18], [29].

- Horizontal FL: In horizontal FL, different sample datasets with the same feature space over FL participants are utilized for local training (e.g., the banking data of two regional banks have different user groups but the data have the same features).
- Vertical FL (VFL): Vertical FL can be applied to the same sample datasets with different feature spaces (e.g., data of the same user groups served by companies from two different business sectors, such as a bank and an e-commerce company.)
- FTL: The FTL can be applied to different sample datasets with different feature spaces.

C. Hardware Architecture

In FL, the local training tasks are executed on distributed FL participants such as MEC servers, edge devices, etc. These participants typically have limited computational resources, memory footprints, or battery power compared to centralized servers. Thus, specialized hardware design for FL is a promising direction to accelerate local training tasks while using much less energy consumption compared to general-purpose processors. Therefore, to efficiently implement FL, various hardware accelerators have been considered for the hardware architecture of FL [7], [33].

- Graphics Processing Units (GPU)-based accelerators: A GPU is an efficient computing tool that leverages highly efficient matrix-based operations and offers a wide range of hardware choices. GPUs provide high memory capacity, bandwidth, and throughput. However, their power consumption is relatively high, which poses limitations when applying them to large-scale systems or battery-powered devices. Notably, the NVIDIA DGX1 and DGX2 are popular GPU-based accelerators that provide accelerated deep learning performance. It is worth mentioning that research efforts have been made to enhance the energy efficiency of GPU-based FL processes [34].
- Field-programmable Gate Array (FPGAs)-based accelerators: FPGAs provide energy-efficient computing and offer reduced costs for large-scale operations. According to Microsoft Research's Catapult Project, FPGAs are estimated to be 10 times more power efficient than GPUs. However, utilizing FPGAs requires a significant amount of storage, external memory, bandwidth, and computational resources capable of handling billions of operations per second. In the context of FL, FL participants need to have sufficient storage capacity for local datasets.

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Further research is needed to optimize these limitations and explore ways to address them effectively.

- Application-Specific Integrated Circuits (ASIC)-based accelerators: ASIC architectures surpass FPGAs and GPUs in terms of both energy efficiency and computational speed. They also minimize memory transfer. An exemplary ASIC-based accelerator is EdgeTPU, developed by Google, which provides an open and end-to-end infrastructure for implementing AI solutions. However, it should be noted that ASIC development typically involves a longer development cycle and lacks flexibility in handling varying deep learning network designs, which requires further investigations.

D. Networking Requirements

In FL, as illustrated in Fig. 2, the overall delay of the FL process encompasses the local training delay at FL participants from a computing perspective, as well as the uplink transmission delay (i.e., local model upload) and downlink transmission delay (i.e., global model download) from a networking perspective. The downlink and uplink transmission delays are dependent on factors such as the model size, the number of FL participants, their channel conditions, and the overall bandwidth allocated for the FL process. To achieve higher accuracy in a shorter time, involving more FL participants is desirable. However, a large number of FL participants, such as massive connectivity of IoT devices, can lead to increased networking delays due to reduced bandwidth per FL participant and increased interference, among other factors. Consequently, the networking bottleneck is recognized as a significant challenge in FL. For instance, wireless networks have significantly smaller bandwidth compared to wired networks. Therefore, the allocated bandwidth for each FL participant may be limited and further reduced as the number of FL participants increases. This limited network bandwidth not only hampers efficient networking between FL participants and the centralized server but also gives rise to straggler FL participants who fail to share their local updates with the centralized server during communication rounds. Both these issues serve as bottlenecks for the performance of FL deployment [33]. In mobile network management, most distributed MEC servers are interconnected through wired connections, which helps alleviate networking problems due to the high bandwidth of wired connections. However, resource exhaustion caused by control plane overhead in FL still persists, particularly in scenarios where UAV-assisted MEC server environments rely on wireless connections to interconnect MEC servers. Therefore, to meet the low-latency or ultra-low latency requirements of FL, depending on the services involved, various techniques have been studied, including efficient FL participant selection, gradient compression, bandwidth allocation, and over-the-air computation, leveraging the superposition nature of wireless transmission [5], [33], [35].

E. FL Management

Elaborate control and management for computing, networking, and local datasets are required depending on the scenarios considering that FL has various classes of networking structures, synchronization approaches, and data partitioning. FL is generally known to have several properties that distinguish it from a typical distributed optimization problem: non-IID of the local dataset, unbalanced number of datasets compared with the number of FL participants, and the heterogeneity of computing and networking capabilities among FL participants. These factors complicate efforts to increase the efficiency of the FL process and prompted research on various aspects of FL management. Critical challenges facing FL management were summarized in a brief survey [2]. Specifically, scarce radio resources with channels of different qualities and conditions need to be efficiently allocated over FL participants where the FL participants are involved in the FL process via a wireless connection. Moreover, in the case of battery-constrained FL participants, their energy consumption for transmission and computation must be considered for energy-efficient FL training while sustaining strict latency and training efficiency requirements. In the FL process, the convergence time is related to the total number of rounds and the latency per round, which is affected by both computation and communication delay. Correspondingly, to minimize the convergence time to achieve the target accuracy, the tradeoff between latency per round and the number of rounds should be carefully considered to adjust the number of local iterations. For instance, a larger number of local iterations may lead to a smaller number of required rounds by increasing the energy consumption and latency per round while reducing the total number of rounds for convergence. Accordingly, a variety of FL management schemes have been proposed in the literature [36], [37], [38], [39], [40], [41], [42], [43], [44], [48], [49], [53], [54], [55], [56], [62], [32], [45], [46], [47], [50], [51], [57], [63], [64]. We categorized these schemes into three criteria: performance metrics, decision variables, and problem-solving methods, as listed in Table III.

TABLE III
SUMMARY OF RESEARCHES ON FL MANAGEMENT

Crietria	Categories
Optimization variables	• Set of participants for local training [36]–[47] • Data utilization [32], [48]–[52] • Resource allocation [32], [42], [44], [46], [48], [49], [51], [53]–[61] • Other [37], [50], [54], [55], [57], [62]–[65]
Performance metrics	• Energy consumption [37], [44], [46], [48], [50], [51], [53]–[55], [57], [64], [66]–[68] • Learning convergence time [32], [36]–[38], [41]–[43], [48], [49], [51], [54]–[56], [69], [70] • Prediction accuracy [32], [39], [40], [45], [50], [56], [62], [71], [72] • Other [36], [37], [41], [42], [44], [46], [47], [63], [73]
Problem-solving methods	• Convex optimization [44], [46], [49], [53]–[57], [74] • Greedy heuristic [36], [37], [39], [42]–[44], [46], [47], [51], [54], [55], [62] • LP [32], [42], [75] • Iterative methods [48], [57], [76], [77] • Other [38], [40], [41], [41], [50], [63], [64], [78]–[80]

We define an optimization variable (i.e., control or decision variable) as a variable that specifies the way in which to control or manage target resources used for FL. Certain studies considered single decision variable scenarios [36], [38], [39], [40], [41], [43], [45], [64], whereas others were concerned

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with joint optimization problems with more than two decision variables [32], [37], [42], [44], [46], [48], [49], [50], [51], [53], [54], [55], [56], [57], [62], [63]. The most common types of optimization variables are a set of FL participants for local training [36], [37], [38], [39], [40], [41], [42], [43], [44], [45], [46], [47], data utilization [32], [48], [49], [50], [51], [52], and the configuration of computational/communication resources (e.g., CPU frequency, uplink/downlink capacities, and transmission power) [32], [42], [44], [46], [48], [49], [51], [53], [54], [55], [56], [57], [58], [59], [60], [61]. Apart from the above, other types of decision variables are available to control the BS/server association, IoT device clustering [37], [54], [55], [64], [65], an incentive for participation [50], data augmentation ratio [62], downsampling ratio [62], sub-group leader [62], prediction accuracy provision [57] and so on.

Those above-mentioned optimization variables should be optimized to minimize or maximize the target objective function (i.e., metric) while satisfying the specified constraints. Most previous studies considered the energy consumption of FL participants to alleviate the burden on FL processing [37], [44], [46], [48], [50], [51], [53], [54], [55], [57], [64], [66], [67], [68], convergence time [32], [36], [37], [38], [41], [42], [43], [48], [49], [51], [54], [55], [56], [69], [70], and prediction accuracy [32], [39], [40], [45], [50], [56], [62], [71], [72] as the main performance metrics. For these metrics, tradeoffs frequently exist in many different scenarios. For instance, there might be a tradeoff relationship between i) reducing the energy consumption by lowering the number of FL participants or the amount of data utilization and ii) the prediction accuracy or convergence speed. To balance such a tradeoff relationship, several researchers adopt constraint parameters to specify the performance criteria in their problem formulations [36], [37], [48], [53]. Other researchers designed elaborate cost minimization or utility maximization problems that combine the performance metrics based on the proportion or weighted-sum to balance the tradeoffs [32], [47], [49], [50], [54], [55], [63] between them. In addition, the number of participants [36], [44], [46], [73], data utilization [42], class distinguishability of model [42], fairness among participants [41], and staleness of the model [37] are also considered as optimization objective or constraint.

In general, resource control and management problems are formulated as non-convex optimization problems, for which optimal solutions are difficult to obtain. Therefore, many researchers follow alternative problem-solving strategies by slightly compromising optimality by decomposing or relaxing problems too much more tractable ones, which are then solved by convex optimization [44], [46], [49], [53], [54], [55], [56], [57], [74] or greedy heuristics [36], [37], [39], [42], [43], [44], [46], [47], [51], [54], [55], [62] to obtain still plausible sub-optimal solutions. Other problem-solving strategies that have been adopted are Linear programming (LP) [32], [42], [75], iterative methods [48], [57], [76], [77], game theory [50], multi-armed bandit (MAB) control [38], [41], reinforcement learning (RL) [40], [63], clustering [64], [80], a trust-region-based successive convex approximation method (SCA) [78], an adaptive delayed-SGD method (AD-SGD) [79], and Lyapunov optimization [41].

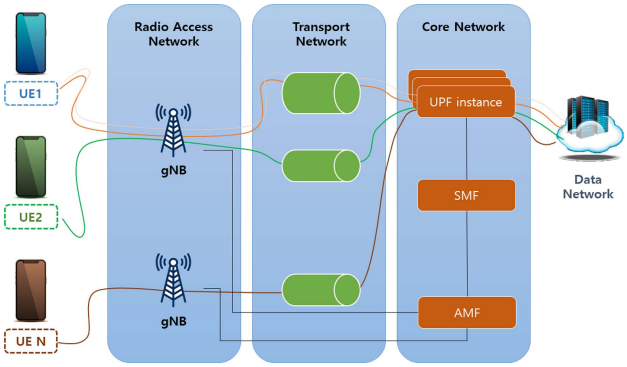


Fig. 4. 5G network functions with PDU sessions [81].

III. AI-Empowered Mobile Network Management

In this section, we present state-of-the-art developments in the field of AI-empowered mobile network management.

A. Background Information About 5G and Beyond Networks

The connection between the user equipment (UE) and the data network in 5G requires the establishment of Protocol Data Unit (PDU) sessions where 5G allows each UE to establish multiple PDU sessions simultaneously to support various services with different QoS requirements at the same time. Here, each PDU session can also belong to a different network slice. As shown in Fig. 4, the UE first connects to a next-generation NodeB (gNB) in the access network to initiate this PDU session. Then, the UE establishes the PDU session through a transport network connected to the 5G core networks, which provides the endpoint to the data network. Contrary to the 4G core network, which consists of dedicated types of equipment, the 5G core network consists of a collection of various virtualized NFs that implement a service-based architecture as follows [15], [81], [82]:

- Access and Mobility Management Function (AMF): The AMF conducts access and mobility management for UE and also provides authentication for UE.
- Session Management Function (SMF): The SMF helps establish, modify and release PDU sessions. The SMF also provides inactivity timers to UEs for managing the activation/deactivation of PDU sessions energy-efficiently.
- User Plane Function (UPF): Whereas the AMF and the SMF are control plane functions, the UPF acts as a PDU session anchor to provide a connection for the access network to the 5G core as a UPF. Furthermore, it also provides inspection, routing, and forwarding packets and QoS handling services by applying specific traffic rules.

Here, the service-based architecture comprises NFs that expose services through RESTful APIs. By using these RESTful APIs, NFs can invoke services in other NFs. To expose the services of NFs to service consumers, service producers must register their services with an NF repository function (NRF). Upon request from an NF, the NRF can provide a list of identifiers for suitable service producers by responding to the request [83]. Note that since NFs in 5G

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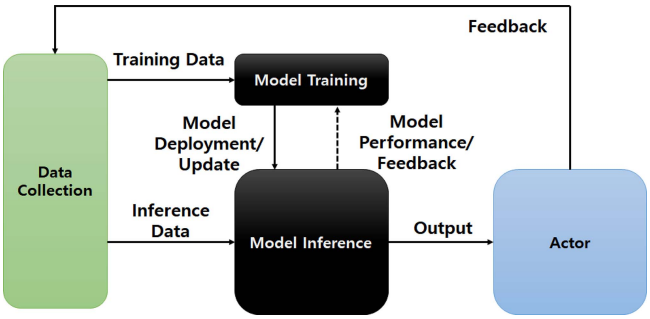


Fig. 5. Functional framework for RAN intelligence [86].

are virtualized functions, these NFs can be virtualized as virtual machines (VMs) or packed into containers that are organized into pods in Kubernetes. Here, a pod is the smallest deployable computing unit in Kubernetes, which contains one or more containers. Depending on the service types, each type of service can be instantiated by a series of NF sets known as network slices, where network slicing has been considered the core of 5G service-based architecture with various service requirements and application scenarios. In other words, the network slice is a chain of NFs with networking (bandwidth) and computing (CPU, RAM) requirements interconnected by a transport network. Various research studies attempted to utilize radio and computational resources for network slicing such that newly emerging services with stringent latency and computational requirements can be supported [84].

Towards 6G networks, it is important to clarify that the official standardization for 6G has not commenced yet. However, we can explore the potential design aspects of 6G by referring to the insights presented in the "International Mobile Telecommunications (IMT) FOR 2030 AND BEYOND" document [85]. This document was published during the Technical Committee of the International Telecommunication Union (ITU-T) 2022 Workshop and outlines the overall objectives for the future development of IMT for 2030 and beyond. Notably, it envisions six usage scenarios, including immersive, massive and hyper-reliable & low-latency communication as extensions of 5G, and new scenarios such as ubiquitous connectivity, integrated AI and communication, and integrated sensing and communication. Furthermore, the capabilities of IMT-2030 highlight AI-related capabilities as one of the key aspects. This indicates that AI-empowered mobile network management will be a crucial technical challenge in designing 6G networks. Additionally, FL can enhance such AI-empowered mobile network management in a more scalable manner by promoting a distributed computing nature.

B. Standard Activities for AI-Empowered Mobile Network Management

1) Access Network: On the access network side, the pioneering O-radio access network (RAN) alliance, which introduces algorithmic and hardware-independent RAN deployment, was established in 2018 by leading operators with the vision of developing open specifications for an open and efficient access network that leverages AI for automating different NFs and reducing operating expenses (OPEX). The O-RAN architecture comprises two essential components: the non-realtime RAN intelligent controller (non-RT RIC) and the near real-time RAN intelligent controller (near-RT RIC). The non-RT RIC is responsible for making non-real-time decisions, including policy management and analytics. These decisions can take longer than one second and are executed through the O1 interface. On the other hand, the near-RT RIC handles near-real-time control and optimization tasks, such as resource allocation and load balancing. These actions are performed within a short time interval, specifically between 10 milliseconds and 1 second, utilizing the E2 interface. The O-RAN architecture's versatility enables the implementation of AI-enabled RAN, facilitating various use cases of RAN automation. Examples of such applications include traffic steering, quality of experience (QoE) optimization, RAN slice service level agreement (SLA) assurance, and more [87]. In particular, ML-based network management and automation, known as xAPPs, running on RAN intelligent controllers (i.e., near-RT RIC) have been designed in the O-RAN architecture aiming for collecting near real-time RAN information to provide value added services. The authors [88] mentioned that these xAPPs still have limitations in that large-scale datasets and an experimental testing infrastructure are not yet available. They designed the first publicly available large-scale O-RAN testing framework with software-defined radios-in-the-loop. In this testbed, deep reinforcement learning (DRL)-based control of RAN slicing, scheduling, and online model training was implemented and evaluated.

Moreover, 3GPP specification series 37 provides RAN-centric data collection and utilization for long-term evolution (LTE) and New Radio and on next-generation self-organizing networks (SON) for 5G networks and beyond [89]. In particular, [86] demonstrates that the AI/ML functionality and the following use cases are presently under investigation within 3GPP. It is important to note that the specific AI/ML algorithms and models for these use cases are considered implementation-specific and are beyond the scope of this standard specification. Consequently, the study mentioned in [86] concentrates on the AI/ML functionality for the specified use cases, along with the potential inputs and outputs that can be collected and generated at the access network. Fig. 5 depicts the functional framework for RAN intelligence specified in [86]. The data collection function serves as the provider of training/inference data to the model training and inference functions. It acts as the input data source for these processes. However, it is important to note that the data preparation process, including tasks such as data pre-processing, cleaning, formatting, and transmission, is performed separately and is not part of this function. Once the model is trained or updated, it can be delivered to the model inference function through the model deployment/update procedure. The model inference function is responsible for monitoring the performance of the AI/ML model and providing feedback to the model training function, facilitating continuous improvement. Subsequently, the actor receives the output from the model inference function and triggers corresponding actions to be taken by other entities or itself. The actor function plays a crucial role in the feedback

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loop, as it provides feedback to the data collection function, which is essential for deriving training data, inference data, and other relevant inputs. Based on this functional framework, the following use cases are proposed:

- AI/ML-based network energy saving: Cell activation/deactivation is one of the methods to achieve energy savings in the RAN by leveraging traffic offloading in a layered structure. In this context, UE location information and UE measurement reports are considered as input data from the UE. Additionally, input data for training can be also obtained from the local node, including UE mobility, trajectory prediction, current/predicted energy efficiency, and resource status
- AI/ML-based load balancing: The main goal of load balancing is to ensure a fair distribution of network traffic across cells and cell areas to alleviate congestion. This optimization process includes adjusting handover parameters and executing handover actions to achieve the desired load distribution and enhance network efficiency. In this context, UE location information, UE mobility history information, and UE measurement reports are considered as input data from the UE. Additionally, input data for training can be also obtained from the local node, including current and predicted own resource status, UE trajectory prediction, current and predicted UE traffic, and predicted resource status information of neighbouring RAN.
- AI/ML-based mobility optimization: Mobility management refers to the strategy used to ensure uninterrupted service during mobility. Its primary aim is to minimize call drops, radio link failures, unnecessary handovers, and ping-pong effects. In the context of future high-frequency networks (e.g., 5G and beyond networks), where the coverage of a BS decreases, frequent handovers become more common, particularly for high-mobility UEs. Similar input data, such as UE information, radio measurement data, and UE mobility history, can be utilized for mobility optimization purposes.

2) Core Network: On the core network side, 3GPP standard support for data analytics is already considered in Rel-16. A data analytics framework, anchored in the new so-called NWDAF located within the 5G core network as a NF following the service-based architecture principles of the 5G core network, has been defined to enhance multiple control-plane functionalities of the network. Moreover, on the operations, administration, and maintenance (OAM) side, a management data analytics function (MDAF) is also being specified by 3GPP to assist with longer-term management aspects of the network [15].

- NWDAF: The NWDAF is being developed by the 3GPP Technical Specification Group Service and System Aspects Working Group 2 (3GPP TSG SA WG2), as shown in Fig. 6 [15], [16]. An analytics service consumer can request specific data analytics from NWDAF, which it provides in the form of field measures and/or predictions. The analytics consumers defined in Rel-16 are 5G core NFs, application functions (AFs), and OAM entities. NWDAF triggers the collection of input data through an exposure framework defined in [15], where the input data sources may again be 5G core NFs, AFs, and/or OAM. It then uses the collected data for training and inference, possibly via an AI engine, but the determination of the inference models is outside the scope of standardization and left open to vendors, which implies that the AI engine may reside outside the NWDAF block. The next standard release (Rel-17) has started studying the required interface standardization to enable NWDAF functional decomposition [90]. Several types of data analytics have been introduced in 3GPP Rel-16, including analytics for network slice and application service experience, NF and network slice load, network performance, and UE aspects (communication, mobility, expected and abnormal behavior), as described in [15]. The ongoing Rel-17 is an expansion of the Rel-16 NWDAF framework and considers many new use cases and key issues, including NWDAF functional decomposition, interaction of multiple NWDAF instances, efficient data collection mechanisms, and support for network slice SLA guarantees [90]. As a practical concern, Release-17 of 3GPP discusses multiple distributed NWDAFs, which depend on specific regions, specific network slices, and/or specific NFs as depicted in Fig. 7. Distributed NWDAFs can collect and analyze data within specific regions and/or NFs, aiming to reduce the network resources used for data collection and addressing security concerns. An example architecture of a 5G core network with distributed NWDAF was proposed [91], where distributed NWDAFs comprise a single root NWDAF in the central cloud and multiple leaf NWDAFs located at the MEC server in a container form. Interfaces among distributed NWDAFs allow them to communicate with each other via the communications module. The learning module in NWDAFs comprises the analytics logical function (AnLF) for the inference process and the model training logical function (MTLF) for the training process.
- MDAF: MDAF provides the analytics service for management data [92]. Although its functional characteristics seem to resemble those of NWDAF, a notable difference exists: MDAF operates in the OAM domain, in contrast to the NWDAF, which resides in the control plane of the 5G core network [93].

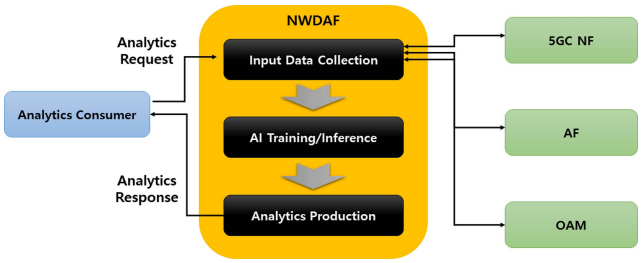


Fig. 6. NWDAF basic operation description [82].

In addition, 3GPP has been developing an network slice management and orchestration framework under the European Telecommunications Standards Institute (ETSI) Management

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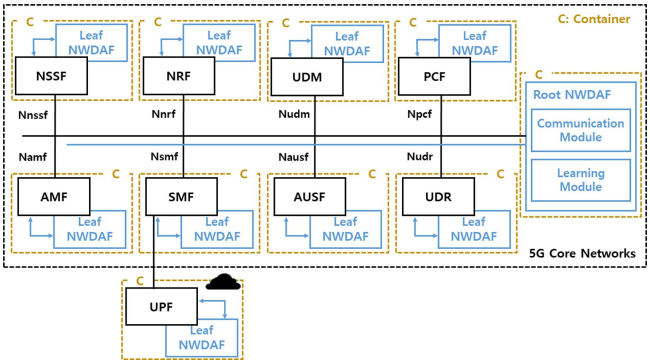


Fig. 7. Example architecture of a 5G core network with distributed NWDAF [91].

TABLE IV
SUMMARY OF RESEARCH ON MOBILE TRAFFIC PREDICTION

Criteria	Categories
Simple method	• EWMA [97]
Parametric methods	• ARIMA [98]–[102] • Markov model [103] • Entropy theory [104] • Covariance function [105] • Alpha-stable model [106]
Non-parametric methods based on deep learning	• LSTM [22], [107]–[109] • DBM [110] • CNN [13], [111] • Transfer learning [112]

and Orchestration and Network Function Virtualization frameworks. The ETSI Experiential Network Intelligence (ENI) group has been studying AI to realize autonomous and cost-effective slice management and orchestration towards AI-empowered network management [94], [95]. Furthermore, ETSI's Zero-touch network & Service Management framework provides a novel horizontal and vertical end-to-end architecture framework for AI-empowered mobile network management as closed-loop automation [96]. These standardization efforts provide initial guidelines for architectural aspects, interfaces, and requirements, while various academic research efforts are geared toward developing specific algorithms based on diverse AI technologies.

C. AI-Empowered Network Load/Status Prediction

According to Zhang et al. [22], mobile traffic prediction schemes can be categorized into three methodologies: simple methods, parametric methods, and non-parametric methods, as listed in Table IV.

Simple methods estimate future traffic flows based on historical averaging [98]. In the case of the exponentially weighted moving average (EWMA), which is one of the most widely adopted methods, we can control the sensitivity for the temporal dynamics of traffic volumes by calibrating the exponential smoothing factor from 0 to 1 [97]. For instance, by assigning the value of 1 to the factor, the EWMA method becomes a naïve predictor that takes the traffic value of the last time slot as the future one. Simple methods are likely to be implemented in a greedy manner, which would probably result in highly efficient runtime execution. Although this type of method can achieve considerable prediction accuracy when the traffic volumes change gradually and slowly, it is less desirable for application in today's complex communication networks where traffic dynamics are highly influenced by a combination of various spatiotemporal factors.

Parametric methods construct probabilistic or statistical prediction models under the assumption that traffic dynamics are generated from the random variables following some specific probability distributions [113]. These methods make predictions by computing or simulating a constructed model. One of the most classical approaches is the auto-regressive integrated moving average (ARIMA) [98]. ARIMA and its variants are known to be capable of characterizing the burstiness and self-similarity hidden in traffic time-series [99], [100], [101], [102]. Xu et al. [101] showed that temporal traffic dynamics can be expressed by the sum of two components, each of which represents regular and random changes in time series, respectively. ARIMA-based approaches deliver excellent prediction performance when the regular change component is large. However, their performance deteriorates as the random change component increases. Apart from ARIMA, a number of probabilistic and statistical utilities such as the Markov model [103], entropy theory [104], covariance function [105], and alpha-stable model [106] are explored for mobile traffic prediction.

Non-parametric methods build prediction models without any assumption about the probability distributions that should be followed. The recent emergence of deep learning technology has resulted in AI-based prediction being brought into the mainstream of non-parametric methods. Because traffic prediction essentially requires the temporal dependence of data in a time-series dataset to be captured, LSTM is often employed, because of its superior ability to recognize sequential patterns [22], [107], [108], [109]. Sevgican et al. [114], under the structure and the protocols of NWDAF, conducted mobile traffic prediction and anomaly detection using various ML techniques. Specifically, they generated a cell-based synthetic dataset for 5G networks based on the fields defined by the 3GPP specifications and used it to train ML models, namely, linear regression, LSTM, and recursive neural networks. With the simulation, they confirmed the performance of neural network algorithms to be superior to that of linear regression for network load prediction, whereas the tree-based gradient boosting algorithm outperformed logistic regression with respect to anomaly detection. Several research groups [22], [107], [108], and [109] combined an autoencoder, multi-task learning, and FL with LSTM, respectively, to characterize not only the temporal dependence but also the spatial dependence between cells or BSs. Others adopted deep belief networks (DBM) [110], CNN [13], [111], and transfer learning [112]. Among them [111] and [112] attempted to model the spatiotemporal dependencies commonly arising in cross-domain datasets to enable city-scale traffic prediction.

The mobile traffic prediction model offers diverse applications in mobile network management. It automatically identifies demand patterns and anticipates future demands,

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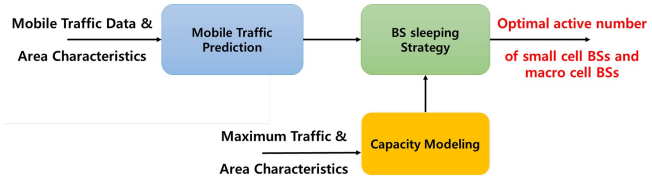


Fig. 8. Example of data-driven BS sleeping control [115].

enabling effective resource management and load balancing. Additionally, it can be utilized for security-related purposes, such as anomaly detection, to enhance network protection [13], [116]. Particularly, Bega et al. [13] represented that the mobile traffic prediction model serves as a valuable tool for network slice resource management, effectively balancing over-provisioning and SLA violations. This use case involves accurately forecasting mobile traffic demands, allowing network operators to optimize resource allocation for each network slice. As another noteworthy example, the mobile traffic prediction model can be also utilized to dynamically optimize a BS's power-on and power-off strategy, commonly referred to as the "BS sleeping strategy". This particular use case has already been addressed in [86] as cell activation/deactivation for AI/ML-based network energy saving. The need for this strategy arises from the common practice of operators deploying an excessive number of BSs to meet peak-hour traffic demands, even when traffic loads are minimal during most of the day. This becomes particularly critical in the context of 5G and beyond networks, characterized by the proliferation of dense and ultra-dense small cells, leading to significant power consumption costs for network operations. Consequently, the selective shutdown of BSs during off-peak hours, based on real-time traffic monitoring, becomes essential for cost-effective network operation. Nevertheless, real-time traffic monitoring for BS sleep control requires excessive control signaling in dense networks. Thus, the application of AI, such as deep learning models, has gained prominence. Data-driven BS sleeping control is now considered one of the emerging use cases [115]. In this data-driven approach, the development of an accurate spatiotemporal mobile traffic prediction model is crucial. For example, spatial features can be predicted using CNN models, while temporal features, encompassing hourly, daily, and weekly periodic data, can be effectively captured using LSTM models. In this context, Zhang et al. [117] presented a multi-view ensemble learning model for mobile traffic load prediction. This model takes into account various perspectives, including i) temporal influence, ii) spatial influence, and iii) event influence, and combines their results in an ensemble manner. The resulting mobile traffic prediction model is then used to formulate an optimized BS sleeping strategy. Similarly, Lin et al. [115] integrated the traffic prediction model with a capacity model that quantifies the maximum traffic capacity supported by macro cell BSs and small cell BSs with varying environmental characteristics. This approach aims to achieve an optimal BS sleeping strategy, as illustrated in Fig. 8. Meanwhile, Wang et al. [118] utilized the mobile traffic prediction model of individual users rather than predicting the traffic for the entire network or individual

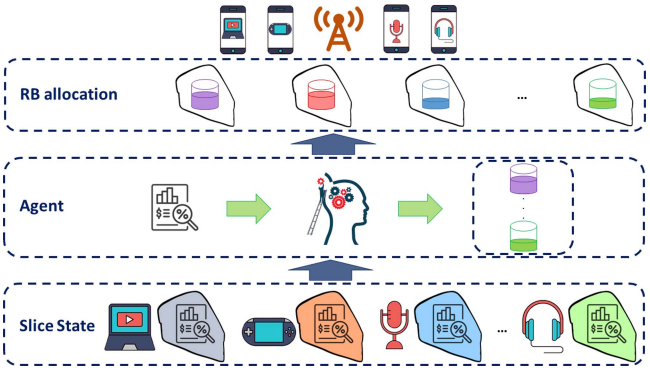


Fig. 9. Overview of DRL-based RAN slicing [120].

BSs. In this work, the goal is to redistribute users from small cell BSs to other BSs and turn idle small-cell BSs into sleep mode, thereby minimizing overall energy consumption. In a similar vein, with the aid of accurate mobile traffic forecasting, Wu et al. [119] also presented a DRL-based BS sleep control.

In addition to the mobile traffic load, a model was designed to achieve a data rate based on either i) only received signal strength indicator (RSSI), the carrier-to-interference ratio, and the terminal velocity, or ii) the previous absolute location of the user terminal by using various ML or deep learning models [121]. Similarly, Lee et al. [122] proposed an LSTM-based mobile throughput prediction model by considering scheduling information (e.g., resource block (RB), transport block size) and received signal reference power (RSRP) as input features. Based on this prediction model, adaptive streaming mechanisms, which adapt the encoding rates with respect to the estimated throughput, were additionally proposed and evaluated. The proposed method outperformed traditional congestion window-based throughput prediction or buffer-status-based algorithms. Moreover, as summarized in [5], to meet the different QoS requirements of vertical services, the design of AI-enabled algorithms for accurate user behavior prediction (e.g., their mobility patterns), channel status estimation, and signal detection has been studied toward the efficient use of radio resources in 6G networks.

D. AI-Empowered Networking Resource Management

AI-based approaches have been studied for implementation in access networks, radio resource management and allocation, and power allocation [120], [124], [125], [126], [127]. Specifically, an RL-based approach was used to achieve power control and rate adaptation in mobile networks [124]. Q-learning-based optimal resource allocation and network access mechanisms in license-assisted access LTE-based heterogeneous networks were proposed [125]. A DRL-based joint user association and resource allocation (i.e., channel allocation) algorithm was proposed to take into account that the formulated problems have non-convex and combinatorial characteristics [126]. Recently, as shown in Fig. 9, a RAN slicing mechanism for slice-level management by using DRL was proposed to allocate radio resources [120]. The DRL algorithm learns the best action, i.e., RB allocation, from various states specified to satisfy the slice requirements with the minimum

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RB allocation using trial and error. The RB allocation is based on multiple defined states, such as throughput requirement, delay requirement, number of allocated RBs, and number of packet arrivals.

A DRL approach was adopted for load balancing within an intracluster group of small BSs, where handover management is automatically conducted based on the historical loads of the BSs [128]. DRL-based handover management was designed to minimize energy consumption resulting from frequent handovers [129]. This approach considers the local information, such as channel gain, angle of direction, required data rate, and the previous UE association vector as the state. This work was extended [130] by incorporating the characteristics of the mmWave environment into a multi-agent DRL process. The A2-threshold-RSRP in the LTE standard is a configuration parameter used to determine the handover decision in practical handover procedures. The threshold value significantly impacts the throughput of cell-edge UEs. This value should be carefully determined by considering many factors, including the traffic load, the number of active users, and channel quality. To address this problem, Chuai et al. [131] proposed an ML or data-driven approach to find the optimal A2-threshold-RSRP, using various factors as features for the model. Polese et al. [132] designed architecture based on an edge controller for mobile networks to dynamically cluster and associate BSs and controllers based on the global mobility patterns of UEs, with the aim of minimizing the number of handover events. The proposed approach was evaluated using real data from hundreds of BSs of a major U.S. operator.

In the context of 5G core network management, a DRL-based solution for managing PDU sessions embedded in the standardized NWDAF was proposed [82]. The solution utilized an inactivity timer-management scheme, which aimed to reduce both UE power consumption and network resource usage. Specifically, the inactivity timer problem is formulated as a Markov decision process (MDP), with the state space represented by a binary vector of the current and past states, the supported traffic type, and the time of day of each PDU session. The DRL model selects appropriate timer values for each PDU session, based on a reward function that balances the power consumption of the UE and the signaling overhead incurred. By using this approach, unnecessary signaling can be minimized while ensuring that the power consumption of the UE is optimized.

E. AI-Empowered Computing Resource Management

Dynamic resource management of clustered computing environments necessitates the scaling of resources for the core network. Two types of scaling methods have been reported in the literature: vertical and horizontal scaling. Vertical scaling involves adjusting the CPU and memory resources allocated to a microservice, whereas horizontal scaling entails scaling in and out of instances for a microservice. The concepts of vertical and horizontal scaling are illustrated in Fig. 10. Recent studies have explored the use of AI techniques for resource scaling. For example, Passas et al. [133] developed an automatic vertical scaling approach for a cloud-native 5G core

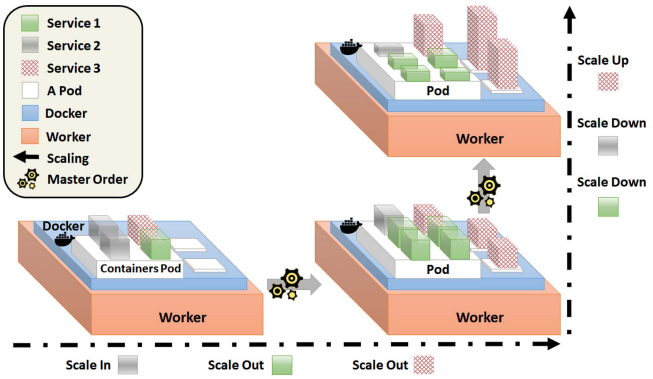


Fig. 10. Example of resource scaling (vertical and horizontal resource scaling) [123].

network, which utilizes a stacked CNN model to estimate the demand pattern. Additionally, the use of DRL for scaling UPF instances packed in Kubernetes containers, shown in Fig. 11, was investigated [81]. This study aimed to minimize the number of pods required to support PDU sessions at the UPF instance while maintaining an acceptable blocking rate. The state information, including the number of active and booting UPF pods, the number of PDU sessions, and an estimated arrival rate, was used to train DRL agents. Similarly, DRL was utilized to address the combined horizontal and vertical scaling problem, as well as service placement decisions for MEC servers. This approach considered multiple Internet of Everything (IoE) services with varying requirements as 6G services [123].

The network-slice placement problem has been the subject of various recent studies, with many utilizing AI techniques to formulate it as a multi-objective integer linear programming problem. This problem takes into account several constraints, such as memory, CPU, disk, bandwidth, and latency, to optimally place NFs on physical nodes. One such study, conducted by Esteves et al. [134], led to the proposal of a DRL-based solution to overcome the convergence problems of integer linear programming. The accuracy was higher than that of heuristics with sub-optimal solutions. To accelerate and stabilize the convergence of DRL techniques, they integrated a heuristic algorithm with the DRL algorithm, following the "Power of two Choices" principle. In another study, Wang et al. [135] designed a DRL-based link-level slice enforcement scheme that meets the critical requirements of network slicing, such as slice isolation, QoS guarantees, and long-term conformance to slicing policies. Finally, Tam et al. [136] investigated DRL-based virtual resource orchestration and appropriate intermediate edge server selection for the intermediate aggregation of local models, considering the heterogeneous IoT data analytics via an FL framework. In their study, the reward of the DRL model is formulated by considering the minimization of the completion time, energy consumption, and round communications between local participants.

The admission control problem for network slices with different priorities, constraints, and network resources was investigated. The complexity of this problem and the required prior knowledge led to the adoption of AI to solve this

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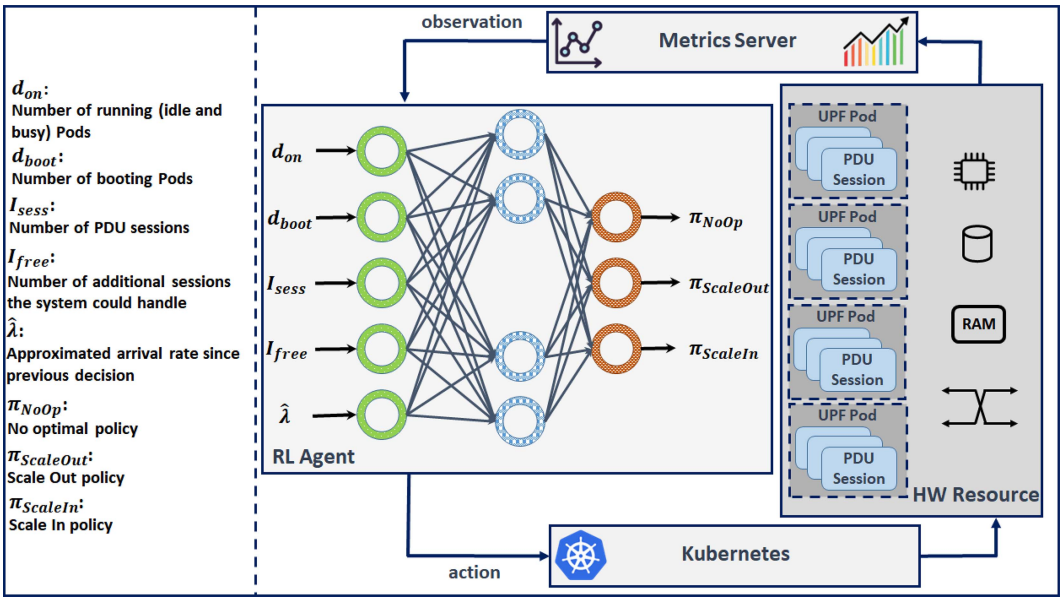


Fig. 11. DRL-based scaling for network management [81].

problem, rather than using various heuristics (e.g., dynamic programming) or convex optimization frameworks. For example, Bega et al. [137] introduced Q-learning-enabled slice admission control, formulated as an MDP that aims to maximize the operator's revenue. When the system is overloaded, the admission control for network slices is jointly managed with the help of resource scaling, as proposed by Dandachi et al. in [138]. To mitigate system overloading, their framework suggests using RL-based automatic scaling resources allocated to low-priority slices while also performing admission control for network slices. Sulaiman et al. [139] developed a joint network slice placement and admission control mechanism using a multi-agent DRL framework to maximize long-term revenue within resource constraints (e.g., bandwidth and computing resources).

F. Industry or Implementation Activities for AI-Empowered Mobile Network Management

Various use cases involving NWDAF and MDAF were implemented in the industry to meet the need for network intelligence. For instance, an NWDAF with an embedded ML model can assist the policy control function (PCF) in determining QoS parameters to fulfill a specific SLA [93]. Another use case is AI-driven traffic prediction, where a prediction model can be integrated into an NWDAF or MDAF to assist an network slice selection function (NSSF) or management function (MF), respectively [13], [93], [140]. In addition, other use cases such as service experience prediction, load analytics, UE behavior, and UE pattern prediction [90] have been implemented by several industry and developer communities. For example, Ericsson provides an NWDAF solution in their 5G platform [141], Nokia launched AVA as Software-as-a-Service (SaaS) for NWDAF [142], and Guavus developed an NWDAF implementation named 5G-IQ, which features vendor-neutrality and interoperability [143]. Moreover, academic and open-source prototypes for NWDAF [144], [145], [146] and MDAF [13], [140] are also available. For instance, an implementation of NWDAF was developed to build an open-source testbed based on free5GC where the free5GC modules are based on the Go language with reference to the NF modules implemented by free5GC [146]. Two research projects investigating the application of AI-empowered mobile network management in the real world are currently in progress [147]. The first project, OREOS, aims to design and implement an end-to-end orchestration platform to support critical services such as vehicle communication, smart cities, and emergency communications that require low latency supported by 5G networks. The OREOS project provides various mechanisms to support the demands of such applications with different priority levels. ONAP, an open-source orchestration and management platform, is used to collect performance, usage, and configuration data from the managed environment, which is then fed to the AI analytic engine to trigger orchestration, such as network slicing instantiation within the 5G network, or to detect anomalies. The second project, POWER, i.e., empowering a digital future, aims to optimize local orchestration tasks such as resource allocation, application placement, and/or migration by utilizing the ML framework. The project supports multiple services provided by Altice, a telecommunication provider in Portugal, such as gaming, set-top boxes, and mobile phones. The project considers hierarchical edge-cloud network architecture to utilize the FL framework for distributed ML.

In summary, we list selected references that consider AI-empowered mobile network management in Table V to summarize the technical aspects including the pros and cons of each approach.

IV. FL-Empowered Mobile Network Management

In this section, we present state-of-the-art developments in the field of FL-empowered mobile network management.

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TABLE V
SUMMARY OF RESEARCH ACTIVITIES ON AI-EMPOWERED MOBILE NETWORK MANAGEMENT

Issue	Approaches	Ref.	Pros	Cons
Deep learning based Network Load/Status Prediction	Deep learning based Mobile traffic prediction	[13], [22], [107]–[112]	1) Deep learning-based non-parametric mobile traffic prediction, including LSTM, DBM, CNN, and transfer learning. (Please see the details in Table IV). 2) Without any assumption about the probability distributions, automatically identifying demand patterns and anticipating future demands for effective resource management and load balancing, optimized BS sleeping strategy, and more.	In real-world implementation, the challenge lies in collecting such data from diverse network entities, leading to privacy concerns and scalability issues, including networking, storage and computing bottlenecks.
	Deep learning based network status and user behavior estimation	[5]	1) Deep learning based network status or user behavior prediction including UE mobility patterns, channel status estimation, and signal detection. 2) Estimation model can be utilized for efficient use of radio resources in 5G and 6G networks.	
	Deep learning based throughput estimation	[121], [122]	1) Deep learning based throughput (i.e., achievable data rate) estimation at mobile devices, including LSTM. 2) Estimated throughput can be applied to adaptive streaming mechanisms to adjust the encoding rates accordingly.	
Networking Resource Management	RL based RAN management	[120], [124]–[127]	1) RL-based management for RAN including power control, user association, and radio resource allocation (e.g., channel and RB allocation). 2) Automation of RAN management without prior traffic model and user distribution by leveraging RL techniques to optimize and enhance the performance of RAN operations.	DRL requires a substantial volume of samples for convergence. However, in the real-world context, gathering diverse samples from multiple network entities poses difficulties due to scalability and privacy concerns.
	DRL-based handover management	[128]–[132]	1) DRL-based handover management for various objects including load balancing, minimization of energy consumption, and reducing handover events. 2) Without any previous mobility models, automation of handover management.	
	DRL-based session management	[82]	1) DRL-based inactivity timer management for PDU sessions aiming for minimizing UE power consumption and network resource usage. 2) Automation of inactivity timers associated with multiple PDU sessions without a previous traffic model and demonstrating the use case of 3GPP NWDAF.	
Computing Resource Management	Deep learning based computing resource scaling for mobile core network	[81], [123], [133]	1) Various deep learning based vertical/horizontal scaling for balancing between resource cost and blocking rate. 2) Automation of vertical/horizontal scaling depending on the dynamic demand to the mobile core without prior knowledge of statistical demand patterns.	In a real-world implementation, the challenge lies in collecting such data from diverse network entities, leading to privacy concerns and scalability issues, including networking, storage, and computing bottlenecks.
	DRL-based network-slice placement	[134]	1) DRL-based network slice placement taking into account various factors including memory, CPU, disk, bandwidth, and latency. 2) Automation of mapping the network slice into physical node by considering dynamically changing environments.	DRL requires a substantial volume of samples for convergence. However, in the real-world context, gathering diverse samples from multiple network entities poses difficulties due to scalability and privacy concerns.
	DRL-based admission control	[139]	1) DRL-based admission control jointly with network slice placement aiming for long-term revenue. 2) Without a previous traffic model, automatic admission control can be achieved.	

A. Distributed NWDAF for FL-Empowered Mobile Network Management

In 5G networks and beyond, because almost all NFs are related to each other, meaningful and valuable analytics can be obtained by aggregating several local models trained by various NFs with different types of input data. This approach is an example of data partitioning in FL. FL architecture in distributed NWDAF was proposed to enable the efficient and secure aggregation of local models in a distributed environment, as shown in Fig. 12 [91]. The architecture includes multiple leaf NWDAFs that gather local datasets from associated NFs (e.g., NRF, AMF, and SMF) and conduct local training using the global model provided by the root NWDAF. The root NWDAFs in the centralized server communicate with multiple leaf NWDAFs to upload and download the model, respectively. Various aggregation methods, such as FedAvg, can be applied to the aggregation module located in the learning module. The root NWDAFs can manage the FL process efficiently using the management module. In this proposed architecture, because the leaf NWDAF is not connected with NFs other than the corresponding NF to which it is connected, the model trained in a specific leaf NWDAF can be accessed only by that leaf NWDAF and the root NWDAF. Thus, the proposed architecture ensures security by preventing any intervention by other NFs or attackers. Using a similar approach, Lee and Shin [148] demonstrated the operation of FL in private 5G networks using the NWDAF similar to Jeon et al. [91]. The researchers considered a distributed NWDAF environment for a private

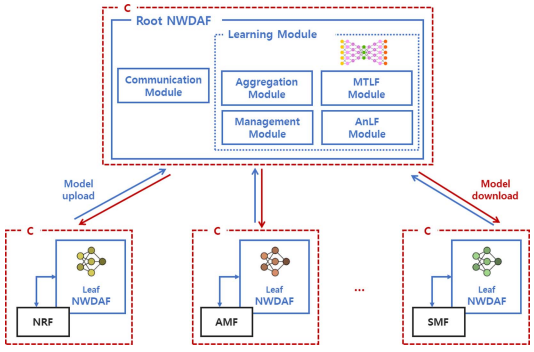


Fig. 12. Operation of FL in distributed NWDAF [91].

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5G network, which is implemented as VMs.[1] To realize secure FL in 5G networks, Isaksson and Norrman [83] integrated the FL framework into the 3GPP NWDAF architecture by considering a multi-party computation protocol for protecting the confidentiality of local updates. Finally, Yuan et al. [150] introduced abnormal behavior detection (e.g., being misused or hijacked) in multiple NWDAFs as one of the critical future research directions to improve 5G network security.

B. FL-Empowered Network Load/Status Prediction

Recently, apart from conventional centralized training, distributed FL-based training approaches for mobile traffic prediction models were proposed to address the shortcomings of the centralized approach, namely the negative impact on the latency, processing costs at the centralized server, and privacy [22]. However, unlike in other research fields (e.g., computer vision and natural language processing), the dataset constructed for the prediction of mobile traffic in wireless networks is neither independent nor identical, which degrades the accuracy of the model and negatively affects the convergence speed [151], [152]. This prompted Zhao et al. to conduct a theoretical study [152] by analyzing the weight divergence between FL with non-IID data and centralized learning with IID data. Based on this theoretical analysis, their findings [152] enabled them to propose a novel data-sharing scheme for FL in wireless networks. Perifanis et al. [153] studied the effectiveness of FL for mobile traffic prediction. Specifically, they utilized datasets from three individual BSs in Spain, ElBorn, LesCorts, and PobleSec,[2] where each dataset comprises a multivariate time-series, i.e., multiple time-dependent variables such as the radio network temporary identifiers (RNTI) count and the number of allocated RBs. Given the measurements of the previous 10-time steps, the authors aim to predict the sizes of the uplink and downlink transport blocks. In their study, they identified the technical challenges of the prediction model trained by the FL process from three perspectives: i) the data distribution skew, ii) the quantity skew, and iii) the temporal skew. Their study revealed that pre-processing techniques such as data cleansing and feature scaling can limit large prediction errors. However, it is noteworthy that convergence issues remain unaddressed. As depicted in Fig. 13, Zhang et al. [22] proposed a novel mobile traffic prediction framework named Dual Attention-Based Federated Learning (FedDA) to boost convergence speeds by promoting the use of multiple MEC servers as FL participants. Specifically, to accommodate statistically heterogeneous datasets collected from different BSs associated with MEC servers, location information, and augmented traffic patterns are utilized to cluster BSs.[3] This work introduced a

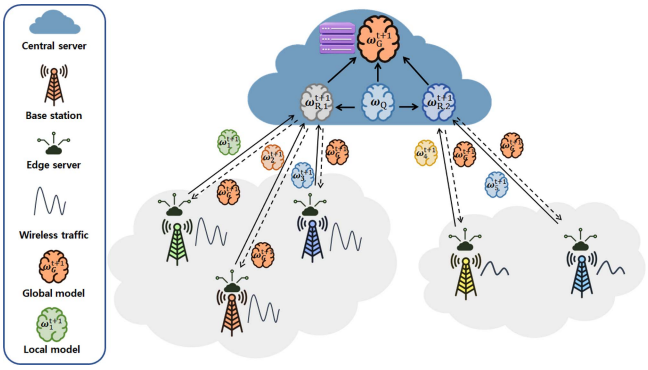


Fig. 13. Example of FL for mobile traffic prediction with five BSs and two clusters [22].

dual attention scheme to efficiently generate the global model after clustering by aggregating the intra- and inter-cluster models. Zhang et al. applied model-agnostic meta-learning to mobile traffic prediction under the FL framework [154]. This approach reduces scratch in the FL training process while obtaining personalized models for geographically distributed heterogeneous datasets. Distance-based weighted model aggregation was also suggested for faster adaptation of an initial model to heterogeneous scenarios from differently located BSs. However, it is important to note that meta-learning is sensitive to hyper-parameters, and thus, optimal choices for these parameters should be determined through a grid search. Recently, by alleviating the straggler impact during mobile traffic model generation due to the heterogeneous computing capabilities of MEC servers, Solat et al. [155] proposed a novel group management scheme by extending clustered FL suggested in [22]. The genetic algorithm was utilized for optimizing i) the number of groups to be created and ii) the group association of the MEC servers to minimize their average idle time and group creation cost. For practical concern, the proposed scheme complies with the NWDAF-distributed framework standardized by the 3GPP specifications. Similarly, Kim et al. [156] proposed a joint MEC server selection and dataset management for reducing training costs, including both training latency and energy consumption in the FL process by balancing accuracy and training cost. To do this, a genetic algorithm was utilized for determining the sub-optimal solutions. However, it is essential to acknowledge the additional complexity introduced by the genetic algorithm. Finally, Li et al. [157] proposed an intra-cluster FL-based model transfer framework for mobile traffic prediction. Here, they assumed that the computing resources of the MEC server include the capabilities of inference and training, respectively, which correspond to MTLF and AnLF in NWDAF. Specifically, they clustered FL participants based on regional and statistical features, and within this framework, manually defined features were found to be simple and effective. Transfer learning was utilized to enable MEC servers with limited computing power to utilize models trained by MEC servers with sufficient computing power. However, it is important to note that their approach lacks the utilization of other multidimensional data, such as regional population

[1] A distributed NWDAF environment and private 5G networks are configured using Free5GC, a well-known 5G open source project [149].

[2] Federated Traffic Prediction for 5G and Beyond Challenge: https://supercom.cttc.es/index.php/ai-challenge-2022.

[3] To address privacy concerns, instead of sending raw mobile traffic data, each MEC sends its augmented data and geographic location information. The augmented data are obtained by computing the statistical average value for each time point following the steps described in Section IV-A [22]. This ensures that the augmented dataset is smaller than the raw dataset, yet the augmented data are highly similar to the original data.

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density and emergency occurrence information, as well as some additional traffic features obtained through AI. Other researchers [158], [159] discovered that network slicing allows multiple mobile virtual network operators (MVNO) to offer various services where those services run on the same network infrastructure owned by an infrastructure provider. Here, MVNOs are required to estimate the future traffic demand in advance for the pre-allocation/reconfiguration of the resources at the BS while preserving privacy from both the business and user-level point of view. To satisfy these requirements, Phyu et al. [159] designed the federated proximal long short-term memory (FPLSTM) framework by incorporating a proximal term to mitigate the impact of local models on the global model. It should be noted that the selection of the proximal term requires careful consideration due to its sensitivity. Then, after carefully considering large-scale cellular networks, the scope of this research was expanded to incorporate two distinct clustering mechanisms. Specifically, information-based clustering utilizing traffic trends of slices and geographical coordinates of BSs, and random clustering were implemented [160]. These enhancements contributed to improved accuracy, scalability, and robustness.

Besides developing a model for predicting mobile traffic, there has also been research focused on using the FL framework to forecast data volume and various network status. Reference [161] proposed a BiLSTM-based FL framework to predict the data volume for computational offloading over mobile devices. The optimal computation resource allocation and offloading decision at each mobile device can be determined based on the prediction model obtained using the FL framework. To detect personally identifiable information, tracking, ad requests, malware, or other activities by inspecting mobile traffic transmitted out of mobile devices, the authors of [162] proposed an FL approach for mobile packet classification by training classifiers in a distributed manner in which features extracted from mobile HTTP packets are learnt. With the advancement of AI, predicting the Key Performance Indicator (KPI) of running network slices, which consist of NFs, has been studied to proactively react to any service degradation of running network slices [163], [164], [165], [166], [167], [168], [169], [170]. In [164], an FL was applied to build a learning model to predict the service-level KPI of NFs (e.g., AMF in 5G). The in-slice manager located at the NFs was in charge of monitoring the service-level KPI and training the local FL model because such service-oriented KPIs are privacy-sensitive data and critical information on service performance should not be disclosed. Particularly, OpenAirInterface is utilized to generate simulated datasets of NFs, which consist of four features: Slice ID, number of CPU resources, number of memory resources, number of attached users as input data, and average duration of user attachment as output data. Moreover, considering that the amount of training data for producing an excellent CSI prediction model is insufficient for some small BSs, [171] proposed an FL-based CSI prediction model in which the global model is trained at the macro BS by aggregating the local models delivered from the small BSs. By adopting the FL framework to generate the CSI prediction model, the considerable communication overhead associated with sending raw CSI data can be alleviated. Reference [172] designed an FL framework with MEC-based small-cell BSs to form a prediction model for the signal-to-noise ratio aimed at UE-aware resource provisioning. Exploiting the use of the Kolmogorov-Smirnov test for correlative distribution analysis on local data points to capture the extent of the IID nature of datasets, the distribution process of datasets among small-cell BSs is considered to improve computational efficiency. Reference [173] also proposed FL-based channel estimation for conventional and intelligent reflecting surface (IRS)-assisted massive multiple-input multiple-output (MIMO) systems. In this study, a single CNN model was trained on two different datasets for both systems. By addressing the challenges in THz channel estimation, such as i) beam misalignment because of frequency-independent analog beamformers and ii) computational complexity due to the ultra-massive number of antennas to compensate for propagation losses, [174] proposed an FL framework for THz wideband channels and direction-of-arrival estimation as federated multi-task learning.

C. FL-Empowered Networking Resource Management

Efficient bandwidth allocation and power control are crucial tasks in radio resource management for access networks owing to their limited availability. However, optimizing the bandwidth and power control mechanisms is challenging because ultra-dense networks are common, and large configurations of network slices are required. Moreover, sharing infrastructure over multiple operators is common; however, these operators want to avoid sharing operational information because of security concerns. To address this issue, the FL framework can be used for data-driven radio resource management by promoting data sharing among multiple BSs or operators in a distributed manner. For instance, the study of [175] designed a bandwidth allocation problem for the Ultra Reliable Low Latency Communications (URLLC) and enhanced Mobile Broadband (eMBB) UEs of an MVNO as a Markov decision process. In particular, because the data collected by each MVNO in different areas is rarely shared among operators due to privacy issues, the FL framework was adopted to promote cooperation among MVNOs to collaboratively generate a global model for better bandwidth allocation. Specifically, they proposed the federated DRL approach to train the bandwidth allocation model among MVNOs. For practical concern, the proposed work considered an O-RAN architecture. Nevertheless, the paper does not address the challenge of unbalanced and non-IID data across MVNOs, crucial for FL model training and convergence, as highlighted in Section II. In [165], a novel radio resource allocation method for RAN slicing was designed under the FL framework using KPI mini-datasets for FL. Particularly, by collecting mini-datasets that contain the hourly traffic of the top-10 over-the-top (OTT) applications, channel quality indicator (CQI), and MIMO full-rank usage as input features, the model for predicting i) the number of occupied downlinks, ii) physical RBs, and iii) CPU load as

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a resource allocation vector was trained offline via the FL framework while fulfilling SLA constraints. To meet the SLA constraints in FL training, a proxy-Lagrangian framework is utilized for local optimization tasks. However, the complexity of this framework could pose challenges for deployment in real-world systems. There have been studies to investigate the possibility of the FL framework under O-RAN to provide resource management for 5G slicing services [176], [177]. Singh and Nguyen [176] specified RAN intelligent controllers based FL framework for slice-level resource management under the O-RAN architecture. The process involves collecting and storing slice operational data into distributed databases through the E2 interface. Subsequently, this data is transmitted to the near RT-RICs deployed on distributed MEC servers for local training. Meanwhile, the non-RT-RIC on the centralized server performs global model updates by aggregating the local model updates received from the near RT-RICs via the A1 interface. Finally, in this paper, joint optimal resource allocation and selection of FL participants, which involves identifying representative instances of near-RT-RICs, was formulated and solved. Moreover, in [177], under the O-RAN architecture, to minimize the delay of URLLC slices and maximize the throughput for eMBB slices, joint power control and radio resource allocation functions (i.e., xAPPs in the O-RAN) were proposed based on the federated DRL. Particularly, two xAPPs, which are ML-based network management and automation running on RAN intelligent controllers, such as a power control xAPP and a hierarchical radio resource allocation xAPP, are considered where the local models for the FL are trained at each xAPP agent and serve as the input for predicting a joint Q-table. Furthermore, for industrial IoT services, in [178], an FL-based DRL scheme was proposed to enable dynamic network management and resource allocation with differentiated QoS requirements of industrial IoT services. Particularly, according to the slices' QoS requirements, the proposed network management and resource allocation were mapped to the spreading factor and transmission power. Correspondingly, a multi-agent deep Q-learning-based dynamic slice transmission power and spreading factor adjustment process was designed to maximize self-QoS in terms of throughput and delay, where multi-agent self-models are locally trained over distributed agents and aggregated at a central server to exploit shared experiences between agents. For practicality, they considered an network slicing architecture based on SDN and NFV tailored for industrial IoT, with a specific focus on Long Range (LoRa) slicing. However, the studies by [177], [178] also lack specific FL management strategies to address convergence challenges caused by unbalanced and non-IID data among FL participants.

As ultra-dense networks are expected to experience more frequent handover events, efficient handover management in access networks is critical for load balancing and efficient radio resource use. Recently, extensive research has been conducted using FL to design distributed data-driven algorithms for the handover management of multiple BSs. Particularly, the authors of [179] proposed a federated DRL-based UE access control aimed at improving throughput while reducing handoff costs, where UEs need to determine the UE network slice-BS three-layer association to meet the QoS for the required service. In this study, a horizontal FL is adopted for parameter aggregation on the BS, whereas the VFL is adopted for network slice/BS pair selection aggregation on the encrypted party. The work introduced Shapley values for efficient VFL aggregation, reducing communication overhead during FL model updates. However, this approach faces challenges when applied to UEs with limited battery and computing capabilities, as some UEs may struggle with FL task processing. Additionally, due to the heterogeneity of UEs, datasets collected can be unbalanced and non-IID, which necessitates addressing FL management strategies to tackle convergence issues. Similarly, in [180], under O-RAN, to solve the challenging problems of load balancing and frequent handovers in the massive BS deployment under the O-RAN, a federated DRL-based scheme was designed with a centralized global model server installed in the RIC to update the deep Q-network (DQN) parameters, where each UE trains two DQNs using its observations (i.e., at previous time slot, BS access indicator, a vector of the RB allocation, a vector of RSSI in different RBs and different BSs) and makes access decisions (i.e., selecting proper BS and RB to access) based on its DQNs to maximize the throughput and avoid frequent handovers. The trained DQN parameters at each UE were considered a local model and transmitted to a global model server. In this work, an upper-confidence bound-based UE selection mechanism for FL participation is proposed to reduce the signaling overhead between UEs and the global model server for model updates. To mitigate convergence issues arising from non-IID data, a dueling structure was introduced to decompose the DQN parameters for each UE. However, similar to [179], this proposed framework also places a computational burden on UEs for FL participation. Addressing this, such as through incentivization strategies, optimal selection of UEs, or burden reduction, remains an open challenge in this problem.

D. FL-Empowered Computing Resource Management

Various studies have been explored for utilizing the FL framework in automating the scaling and provisioning of computing resources. In [168], FL-based vertical autoscaling in multidomain networks was proposed, where the problem of autoscaling was designed as a time-series forecasting problem. Thus, the FL-based model is trained to predict the future number of NF instances based on the expected traffic demand for autoscaling. The study evaluated performance by analyzing both sequential and parallel aggregation methods in various deep learning algorithms. Additionally, they presented a prototype of AI-driven Kubernetes orchestration. While comprehensive evaluations were conducted on FL-based auto-scaling, efficient combination of sequential and parallel aggregation, and FL management addressing convergence challenges arising from unbalanced and non-IID data across multidomain networks should be further investigated as future research. The authors of [169] extended this work by providing a fully DFL architecture over distributed NFs without a

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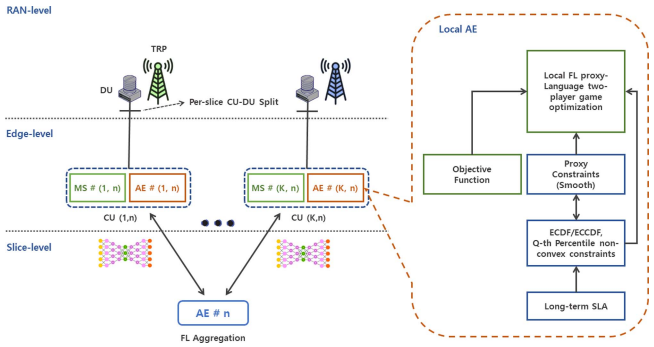


Fig. 14. FL-based decentralized analytics engine for slice n [181].

centralized server for the training model. This approach potentially lowers communication overhead compared to the CFL. Their work also considered a network architecture compliant with ETSI NFV standards. However, achieving acceptable convergence in the DFL is more challenging than in the CFL, necessitating the development of more sophisticated aggregation methods to effectively tackle these convergence issues. In [181], a decentralized AI analytics engine was built over edge servers to improve energy efficiency using the FL framework for resource analysis in network slicing. The architecture, illustrated in Fig. 14, involves an analytics engine (AE) with a 6G tailored central unit (CU)-distributed unit (DU) functional split per slice. Each CU operates as a virtualized NF on a commodity edge server and collects slice-level RAN-related KPI data through a local MS. The AI is also implemented to enable AI-enabled slice resource provisioning using these data. To leverage the FL framework, the AE instances of each slice conducted local training with their datasets and shared their local models with the AE of an aggregation server. Moreover, this paper includes detailed discussions on the application of federated DRL and ETSI Zero-Touch Network and Service Management. Similar to [165], a proxy-Lagrangian framework is used for local optimization tasks to satisfy the SLA constraints in FL training. However, as discussed, the complexity of this framework may present challenges when deploying in real-world systems.

In [182], the placement strategy for NFs was determined based on the forecasted service demands using the FL framework. In this framework, each end client requests an network slice service associated with different classes such as eMBB, URLLC, and massive Machine Type Communications (mMTC). The service provider acts as an network slice controller and provides a network slice to achieve maximum revenue through efficient placement of the NFs. Here, the distributed end clients have their own local historical datasets associated with a specified service category. The dataset consists of a series of recently requested NFs used for local training to obtain a model that predicts the next NF to be requested. The FL framework is then used to perform iterative interworking between the network slice controller and the end clients. A matching theory-based NF placement strategy was proposed based on a global prediction model and the perceived QoE. However, this approach encounters difficulties with UEs that have limited battery and computing power, as some may struggle to process FL tasks. Moreover, the heterogeneity of UEs can lead to the collection of unbalanced and non-IID datasets. This situation requires the development of effective FL management strategies to address convergence challenges.

Regarding DRL-based network slice orchestration, [183] addressed the cold-start problem of DRL for network slice orchestration across multiple service providers. The cold-start problem refers to the fact that a large number of samples are required to achieve DRL model convergence, leading to significant orchestration costs [184]. To overcome this challenge, Dong et al. [183] proposed a federated meta-reinforcement learning (FedMRL) approach for network slice orchestration. This approach includes performing tasks such as VM deployment, flow scheduling, and bandwidth allocation while enabling fast adaptation and good generalization in dynamic 6G network environments. In addition, the proposed method addresses privacy concerns during training because it is performed on the network slices of multiple service providers. By addressing this cold-start problem, the proposed approach can help reduce orchestration costs and improve the efficiency of network slice orchestration. However, it is important to note, as discussed, that meta-learning's effectiveness heavily depends on hyper-parameter settings. Therefore, identifying the optimal values for these parameters typically requires conducting a grid search.

Given that the performance of network slices relies on the normal operation of VMs, detecting anomalies in VMs is critical for ensuring the quality of network slice services. Anomalies can be defined as any performance degradation that deviates from normal behavior, and detecting them quickly is important to address security concerns. The performance and security of network slices can be improved by proactively detecting and addressing anomalies in VMs. To address this issue, [185] proposed a hierarchical anomaly-detection mechanism using the FL framework. The proposed mechanism utilizes two stages of the FL framework to efficiently detect anomalies. In the first stage, an anomaly detector is associated with an aggregation server at the management domain level. This detector uses a simple ML model with a relatively small database. In the second stage, an anomaly detector is associated with an aggregation server at the end-to-end management-domain level. This detector utilizes a more complex ML model and a relatively larger database than that at the first stage. By utilizing this hierarchical approach, the proposed mechanism can effectively detect anomalies in network traffic while minimizing the required computational resources. Any flow detected as an anomaly during the two stages was immediately removed. As practical considerations, they adopted ETSI Zero-Touch Network and Service Management as an underlying system model. However, it is well-known that data collected from VMs tends to be high-dimensional and frequently imbalanced. Furthermore, obtaining abnormal data proves to be difficult. These factors pose significant challenges in developing an effective FL model. To alleviate this concern, Wang et al. [186] proposed a three-tier distributed anomaly-detection framework based on FL to identify abnormal behavior in VMs. Considering the high-dimensional

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and imbalanced data features in VMs, the FL framework was combined with a new multi-discriminator bidirectional Wasserstein Generative adversarial network (GAN)-gradient penalty algorithm. This algorithm is designed to learn the normal data distribution from high-dimensional metric datasets spread over multiple VM monitors, where the resource metrics are related to CPU usage, memory consumption, disk read/write, and network input/output. Nevertheless, the complexity of the multi-discriminator bidirectional Wasserstein GAN-gradient penalty algorithm may make the approach challenging to implement.

In 5G networks and beyond, to reduce the service latency, caching content over MEC servers, which are associated with BSs, has been widely studied. Owing to the limited storage and computing resources of MEC servers, efficient cache management has become an important research direction in AI-empowered mobile network management. Considering the isolated dataset of MEC servers, some studies have adopted the FL framework to share AI models among MEC servers in a collaborative manner so that data-driven cache management over MEC servers can be achieved in a distributed manner [187], [188], [189], [190], [191], [192]. Particularly, in [187], the focus is on employing FL for edge caching in mobile IoT networks. This approach aims to predict the future popularity of content, optimizing its placement to enhance the QoS while considering the dynamic mobility patterns of UEs. Specifically, the research utilized an SDN-controller as the global aggregation point for the FL. However, the proposed method imposed a computational burden on the UEs due to the FL process. Furthermore, the heterogeneity of UEs can result in an unbalanced and non-IID dataset. Thus, it necessitates the design of efficient FL management strategies, specifically designed to tackle the convergence issues. The article in [189] provided a DRL-based FL framework over MEC servers to jointly optimize content caching and offloading management. By intelligently utilizing the collaboration between UEs and MEC servers to exchange model parameters, dynamic system-level optimization and application-level enhancement were achieved. In [190], federated DRL-based cooperative caching was proposed to optimize the overall cache hit ratio at the MEC server while reducing the communication and computation overhead for training. Particularly, in this study, an FL-based parameter-sharing mechanism between the knowledge database and the MEC server was introduced. The work of [192] proposed a cache replacement mechanism based on federated DRL in hierarchical mobile edge-cloud systems. This mechanism aims to achieve a balance among similarity cost, latency cost, and cache hit cost. It integrated the recommender system that supports both direct hits and soft hits, enhancing the overall effectiveness of the caching strategy. They employed an actor-critic DRL model, which is notably sensitive to the selection of hyperparameters and thus, optimal choices for these parameters should be carefully determined. For 5G and beyond networks, UAVs complement terrestrial mobile networks by extending the network coverage in ultra-dense tiny cell scenarios, such as flying BSs [193], [194]. As shown in Fig. 15, the authors of [195] designed a distributed heterogeneous computing platform

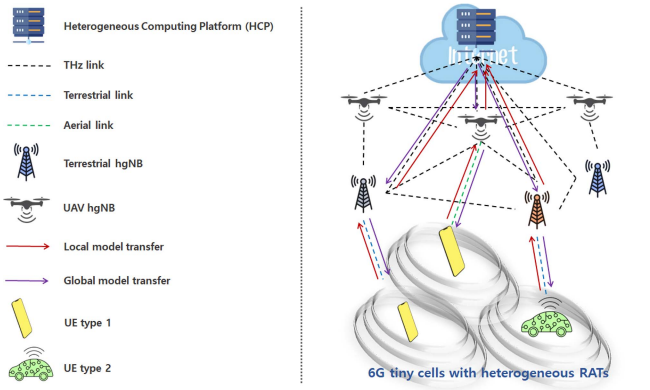


Fig. 15. FL-based cache management based on UAV and terrestrial mobile networks [195].

(HCP) across the UAVs and terrestrial BSs by utilizing their caching and cooperative communication capabilities where the SDN controller is implemented based on the HCP. In this study, under collaboration among UEs, UAVs/BSs, and HCP in this architecture, an FL-based content caching placement prediction model was proposed by jointly considering traffic distribution, UE mobility, and localized content popularity.

E. Architecture or Implementation Activities for FL-Empowered Mobile Network Management

Considering virtualization perspectives in the MEC environment, there are few studies on FL-based architectures for FL-empowered mobile network management [83], [91], [150], [168]. The study in [168] provided a prototype of AI-driven Kubernetes orchestration in MEC-enabled mobile networks. Particularly, in the implementation of a prototype to exchange data (centralized learning) and model updates (FL) between the central server (master) and local MEC hosts, which are capable of hosting MEC applications (workers), Web socket server-client connections are utilized. A distributed streaming platform (e.g., Apache Kafka) is another option for implementing real-time data pipelines instead of WebSocket connections for practical deployment. Moreover, to realize predictive autonomic capabilities in Kubernetes orchestration, the Kubernetes architecture is extended to include a monitor, analyze, plan, and execute (MAPE) closed control loop, which is as follows:

- Monitor component: it collects operational data (e.g., CPU) from all deployed Pods in the worker nodes in the core network, either through a centralized or a distributed manner
- Analyze component: it utilizes the collected data to perform data analytics for predictive autoscaling. It decides the necessity of an adaptation for the deployed pods.
- Plan component: it defines an adaptation plan if it is required based on the result of the analysis component.
- Execute component: it executes the adaptation plan defined by the plan component. It is located at the Kubernetes master.

To efficiently adopt the FL framework in the system, in the work of [166], which is an extension of the work

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in [165] using a similar dataset for the FL in [165], an SLA-driven stochastic FL participant selection policy was proposed to optimize the FL computation time and the underlying resource consumption by selecting only a subset of active FL participants in each FL round. To emulate the virtualization environment widely used for cloud-native deployments, they provided implementation details of the Docker compose-based FL framework, which can be supported by the container orchestration engines specified in the ETSI NFV IFA 029. Another concern regarding securely deploying the FL framework for 5G networks and beyond without the intervention of centralized model aggregation nodes is that some approaches have adopted blockchain technology as an underlying platform to sustain the FL framework [196], [197], [198], [199]. In FL, the centralized model aggregation can be easily exposed to server attacks, resulting in a single point of failure. This greatly biases the global model as intended by attackers. To address this issue and induce trust in FL communication for mobile network management, blockchain technology has emerged as a promising framework. In a blockchain, data is stored as blocks linked to an immutable chain, and the ledger state is communicated to all nodes in the blockchain network. Consequently, blockchain eliminates the need for third-party security management [199]. An exemplary adoption of blockchain technology for mobile network management can be observed in UAV orchestration, which has gained significant attention. In this context, the holistic integration of blockchain-leveraged FL-assisted UAVs ensures the utmost level of data privacy, trusted exchange, and traceable access under continuous local updates due to topology change with diverse UAV mobility, models, and network constraints. As another example, [197] proposed a blockchain-based FL architecture called Skunk's FL system with a zero-trust security model. They eliminated the role of centralized model aggregation nodes by suggesting a blockchain-based DFL environment. In this architecture, multiple FL participants over different network slices can participate in FL, and individual FL participants can share local model parameters using the blockchain's smart contract. To use this architecture, building an ML model throughout the network slices to detect attacks from IoT networks in a 5G-enabled network-sliced environment was introduced as a use case. Therefore, various benefits of utilizing blockchain in FL, such as traceability, decentralization, scalability, immutability, non-reusability, enhanced security, and privacy, can be leveraged to design AI-empowered mobile network management in a secure manner.

In summary, we list selected references of FL-empowered mobile network management in Table VI - Table IX to summarize the technical aspects including the pros and cons of each approach as well as practical perspectives.

V. Lesson Learned

In this section, we summarize the key lessons learned from this survey, which thus provide an overall picture of the current research of FL-empowered mobile network management based on its connection to the existing challenges of AI-empowered mobile network management.

A. Lessons Learned From AI-Empowered Mobile Network Management

1) Standards Activities: AI has emerged as a pivotal facilitator for achieving end-to-end network automation in all aspects of 5G networks, including the access network, core network, and OAM management systems. Consequently, our survey reveals significant standardization activities aimed at defining the specifications for AI-empowered mobile network management, spanning from access to core. Within the 3GPP specifications, there has been an extensive exploration of RAN-centric data collection utilization. A detailed functional framework for RAN intelligence has been established, accompanied by comprehensive use cases for RAN-level data collection including AI/ML-based network energy saving, load balancing, and mobility optimization. On the core network side, 3GPP specifications incorporate the NWDAF and MDAF, providing data analytics frameworks for the control plane of the 5G core network and OAM domain, respectively. Additionally, ongoing development within 3GPP encompasses an network slice management and orchestration framework under the ETSI. Moreover, the ETSI's Zero-touch network & Service Management framework has introduced a pioneering horizontal and vertical end-to-end architecture for AI-empowered mobile network management, focusing on closed-loop automation. In conclusion, these standardization efforts offer valuable initial guidelines concerning architectural aspects, interfaces, use cases, and requirements for the collection and analysis of diverse data generated within the network domain, thus paving the way for network management automation.

2) Network Load/Status Prediction: In the pursuit of network load/status prediction, significant efforts have been dedicated to developing deep learning-based models including mobile traffic, network status, user behavior, or throughput prediction. Primarily, supervised learning techniques such as LSTM, CNN, and others have been extensively employed to capture spatial and temporal patterns without relying on the statistical distribution of traffic models. The research contributions in AI-empowered network load/status prediction present valuable potential for various applications. These include resource management, enhancing load balancing, implementing optimized BS sleeping strategies, advancing adaptive streaming mechanisms, and others, as previously discussed. However, a common assumption among these contributions is the availability of significant training data, which is crucial for achieving higher prediction accuracy. In a real-world implementation, the challenge lies in collecting such data from diverse network entities, leading to privacy concerns and scalability issues, including networking, storage, and computing bottlenecks.

3) Networking Resource Management: AI-based approaches have been studied for designing radio resource management and allocation, power allocation, handover, and session management. Specifically, in this field, a DRL-based approach was widely used. This is due to the fact that DRL is a great tool to design the decision-making algorithm in complex and dynamic environments as discussed in the various studies. Specifically, to handle the massive state and action

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TABLE VI
SUMMARY OF RESEARCH ACTIVITIES ON FL-EMPOWERED NETWORK LOAD/STATUS PREDICTION

Issue	Approaches	Ref.	FL methods	FL management	Pros	Cons	Etc.
FL-based Network Load/Status Prediction	FL-based mobile traffic prediction	[153]	CFL	Pre-processing techniques, including data cleansing and feature scaling.	Limit large prediction errors.	Convergence issues are not addressed.	Identified technical challenges in FL-based mobile traffic prediction, including i) data distribution skew, ii) quantity skew, and iii) temporal skew.
		[22]	HFL	1) Clustering FL participants based on location information and augmented traffic patterns. 2) Dual attention for efficient aggregation of the intra- and inter-cluster models.	Enhanced convergence.	1) Increased complexity in aggregation mechanisms for intra- and inter-cluster models. 2) Lack of consideration for heterogeneous computing capabilities.	-
		[160]	HFL	1) Clustered FL participants using i) traffic trends of slices and geographical coordinates of BSs, and ii) random clustering. 2) Incorporated a proximal term to mitigate the impact of local models on the global model.	Enhanced accuracy, scalability, and robustness.	The selection of the proximal term requires careful consideration due to its sensitivity.	Scenarios involving multiple MVNOs and multiple slices are considered.
		[154]	CFL	1) Federated meta-learning 2) Distance-based weighted model aggregation	Achieve personalized models for a geographically distributed heterogeneous dataset.	Meta-learning is sensitive to hyper-parameters, and thus, optimal choices for these parameters should be determined through a grid search.	-
		[155]	HFL	1) Clustering FL participants based on location information and augmented traffic patterns. 2) Genetic algorithm for efficient group management within the cluster	Enhanced convergence	Additional complexity introduced by the genetic algorithm in creating and managing groups within the cluster.	For practical concern, the proposed FL framework complies with the NWDAF specifications standardized by the 3GPP.
		[156]	CFL	Genetic algorithm for FL participant selection and dataset management	Reduced training latency and energy consumption	Additional complexity introduced by the genetic algorithm for FL participant selection and dataset management.	
		[157]	HFL, FTL	1) Clustered FL participants based on regional and statistical features. 2) Intra-cluster federated learning-based model transfer framework.	1) Manually defined features are simple and effective. 2) Utilized transfer learning to enable MEC servers with limited computing power to use models trained by MEC servers with sufficient computing power	Limitation in not leveraging other multidimensional data, such as regional population density and emergency occurrence information, along with additional traffic features obtained through AI.	
	FL-based data volume prediction	[161]	HFL	-	Applying the default FL framework to intelligent task prediction used for computation offloading	Lack of optimization in FL management.	-
	FL-based network status prediction	[164], [171]–[174]	CFL	-	Applying the default FL framework to various network status predictions.	Lack of optimization in FL management.	-

spaces representing the complex network environment, DRL utilizes deep neural networks to approximate the value, the policy, or both. By defining the agent, environment, state, action, and reward with precision, DRL offers practical and highly effective algorithms capable of addressing critical issues:

- The learning process can be conducted online without stringent requirements on the statistical distribution of the mobility model, traffic model, and other factors.
- Joint optimizations of multiple control variables can be easily combined in one learning model via multi-objective learning.

Despite the advantages of DRL, effectively harnessing its power requires careful consideration of state, action, and reward design. It is also crucial to select appropriate DRL model variants and tune hyperparameters to improve convergence speed, especially in complex environments. As a result, much of the research in this area focuses on designing DRL-based algorithms by formulating the state, action, and reward functions, as well as choosing suitable models to achieve better accuracy while maintaining acceptable convergence speeds. Specifically, in the context of handover management, DRL algorithms aim to minimize energy consumption and reduce handover events. For PDU session management, the focus is on striking a balance between UE power consumption and signaling overhead, and therefore, DRL-based inactivity timer management has been studied. Nevertheless, DRL faces a challenge known as the cold-start problem, where a substantial volume of samples is needed for convergence. However, in the real-world context, gathering diverse samples from multiple

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TABLE VII
SUMMARY OF RESEARCH ACTIVITIES ON FL-EMPOWERED NETWORKING RESOURCE MANAGEMENT

Issue	Approaches	Ref.	FL methods	FL management	Pros	Cons	Etc.
FL-based Networking Resource Management	FL-based bandwidth allocation	[175]	CFL	-	1) Federated DRL-based bandwidth allocation for URLLC and eMBB UEs. 2) Enables multiple MVNOs to create a generalized global model while maintaining data privacy. 3) Adaptive to environment changes due to the shared experience among MVNOs.	The challenge of unbalanced and non-IID data across MVNOs for the FL model's training and convergence.	Design and evaluation of a federated DRL mechanism on an O-RAN architecture for cooperative radio resource allocation among MVNOs.
		[165]	CFL	Proxy-Lagrangian framework for local optimization task	1) Slice-level federated resource allocation under CDF SLA. 2) Downlink physical RB and CPU load SLA violation rates are reduced	1) Complexity of the proxy-Lagrangian framework, which may make the approach challenging to implement. 2) Lack of realistic testbed implementations	-
	FL-based power control	[177]	CFL, VFL	-	Federated DRL-based power control at the BS and RB allocation to each network slice.	The challenge of unbalanced and non-IID data across BS and network slices for the FL model's training and convergence.	Design two xAPPs, namely a power control xAPP and an network slice-based resource allocation xAPP in O-RAN.
		[178]	CFL	-	For industrial IoT service, federated DRL-based joint transmission power and spreading factor adjustment to maximize self-QoS including throughput and delay.	1) The challenge of unbalanced and non-IID data across network slices for the FL model's training and convergence. 2) Lack of realistic testbed implementations.	1) network slicing architecture based on SDN and NFV for Industrial IoT. 2) LoRa slicing is considered.
	FL-based handover management	[179]	CFL, Hybrid FL (Horizontal FL + VFL)	Shapley values for VFL aggregation.	1) Federated DRL-based access control aiming for improving throughput while reducing handoff cost. 2) Communication overhead reduction in FL model updates using Shapley values for efficient VFL aggregation.	1) Increased computational load on UEs due to the federated DRL training process. 2) The challenge of unbalanced and non-IID data across UEs for the FL model's training and convergence.	Leverages VFL for access control of UEs with different services.
		[180]	CFL	1) FL participant selection. 2) Dueling structure introduced to decompose the DQN parameters of each UE for alleviating non-IID data problem	1) Federated DRL-based access control designed to maximize throughput and avoid frequent handovers. 2) Optimal UE selection for FL within the upper confidence bound reduces signaling overheads.	Increased computational load on UEs due to the federated DRL training process.	Consider a centralized global model server in the RIC to update the deep Q-network parameters.

network entities poses difficulties due to scalability and privacy concerns. While some of works consider multi-agent DRL but in this case, a sample still needs to be collected. Unfortunately, this sample collection from multi-agent setups is not easily implementable, especially when dealing with privacy-sensitive information from various operators.

4) Computing Resource Management: Due to networking resource management's similarity to a sequential decision-making problem, research on DRL-based computing resource management has gained significant attention. This research focuses on addressing resource scaling, network slice placement, and admission control problems. In the resource scaling problem, algorithms for vertical/horizontal scaling of virtualized computing resources in the mobile core network have been proposed to strike a balance between resource cost and blocking rate. Additionally, to consider hardware resource constraints such as memory, CPU, disk, and bandwidth, extensive studies have been conducted on network slice placement into physical nodes. Furthermore, admission control, which deals with handling overloaded situations while considering the joint impact of the aforementioned issues, is recognized as a challenging topic. Nevertheless, similar to the limitations of DRL algorithms, concerns related to privacy and scalability need to be addressed for a more practical deployment of DRL-based algorithms.

5) Industry or Implementation Activities for AI-Empowered Mobile Network Management: Various applications of NWDAF and MDAF have been realized in the industry to address the growing demand for enhanced network intelligence. Industry sectors such as Ericsson, Nokia, and Guavus have actively developed and implemented various NWDAF solutions. In the academic domain, a notable project involved creating an open-source testbed using NWDAF, based on free5GC. Additionally, research initiatives such as OREOS and POWER are delving into the potential of AI to refine network orchestration and management, with a specific emphasis on optimizing critical services and local orchestration functions.

Consequently, in response to the previously mentioned common challenges in AI-empowered mobile network

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TABLE VIII
SUMMARY OF RESEARCH ACTIVITIES ON FL-EMPOWERED COMPUTING RESOURCE MANAGEMENT

Issue	Approaches	Ref.	FL methods	FL management	Pros	Cons	Etc.
FL-based Computing Resource Management	FL-based computing resource scaling and provisioning for mobile core network	[168]	CFL	Both sequential and parallel aggregation.	1) FL-based NF autoscaling by predicting the future number of NF instances. 2) Enables multidomain networks to create a generalized global model while maintaining data privacy. 2) Different deep learning algorithms are implemented on the FL framework.	1) The challenge of unbalanced and non-IID data across multidomain networks for the FL model's training and convergence. 2) Not optimized between sequential and parallel aggregation.	Introduce the AI-driven Kubernetes orchestration prototype.
		[169]	DFL	-	1) DFL-based NF placement, migration, and autoscaling by predicting the future processing capacities without a need for a centralized parameter server. 2) Achieves low communication overhead from distributed training.	DFL is sensitive to aggregation methods but lacks consideration of convergence issues.	Consider ETSI compliant NFV network architecture.
		[181]	CFL	Proxy-Lagrangian framework for local optimization task.	1) Energy-efficient slice-level federated computing resource allocation under CDF SLA. 2) Discuss federated DRL applications.	1) Complexity of the proxy-Lagrangian framework, which may make the approach challenging to implement. 2) Lack of realistic testbed implementations.	Discussion ETSI Zero-Touch Network and Service Management.
	FL-based network slice placement	[182]	CFL	-	1) FL-based UE's demand forecasting with different network slices. 2) Based on the prediction model, matching-theory-based NF placement.	1) Increased computational load on UEs due to the FL process. 2) The challenge of unbalanced and non-IID data across UEs for the FL model's training and convergence.	-
	FL-based network slice orchestration	[183]	CFL	Federated meta-learning.	1) Federated meta-reinforcement learning approach for network slice orchestration by promoting collaboration between multiple service providers in a secure manner. 2) Alleviate the cold-start problem in DRL. 3) Include VM deployment, flow scheduling, and bandwidth allocation.	Meta-learning is sensitive to hyper-parameters, and thus, optimal choices for these parameters should be determined through a grid search.	-

management, there has been a growing call for the adoption of FL. This approach has garnered considerable attention and has led to significant recent research developments, which will be summarized in the subsequent subsection.

B. Lessons Learned From FL-Empowered Mobile Network Management

1) Distributed NWDAF: As a use-case of FL in 5G networks, an FL architecture in distributed NWDAF was proposed for efficient and secure aggregation of local models in a distributed environment. The architecture includes multiple leaf NWDAFs that gather local datasets from associated NFs and conduct local training using the global model provided by the root NWDAF. The proposed architecture ensures security by preventing any intervention from other NFs or attackers. Utilizing the distributed NWDAF is suitable for scenarios where multiple MVNOs share infrastructures.

2) Network Load/Status Prediction: There are substantial studies in mobile traffic prediction, which can alleviate the negative impacts of centralized approaches on latency, processing costs, and privacy. However, unlike in other research fields, the collected dataset for mobile traffic prediction in wireless networks is neither independent nor identically distributed, leading to model accuracy degradation and slow convergence speeds. To address these challenges, studies have explored various FL management schemes for mobile traffic prediction under the different types of FL methods (e.g., CFL, HFL, and FTL). A prevalent strategy adopted in this field involves clustering FL participants to enhance convergence. This approach groups participants with similar patterns, mitigating the effects of non-IID data. Location information and traffic patterns are often used as criteria for clustering. Additionally, various optimization techniques have been investigated to improve clustering mechanisms, such as dual attention for efficient aggregation, random clustering, and genetic algorithms for effective group management within clusters. However, these advanced optimization techniques add complexity that needs to be addressed. Most clustered FL studies focus on HFL methods, leveraging their hierarchical aggregation process. Another innovative approach is the use of meta-learning for personalized mobile traffic modeling, though it faces challenges with sensitive hyperparameters that require optimal tuning through methods such as grid search.

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TABLE IX
SUMMARY OF RESEARCH ACTIVITIES ON FL-EMPOWERED COMPUTING RESOURCE MANAGEMENT (CONTINUED)

Issue	Approaches	Ref.	FL methods	FL management	Pros	Cons	Etc.
FL-based Computing Resource Management	FL-based VM anomaly detection	[185]	CFL	-	1) FL-based hierarchical security scheme for anomaly-detection. 2) Simple and complex deep learning model for both the security management domain level and the end-to-end service management domain level.	Lack of consideration of high-dimensional and imbalanced data over VMs.	Align with ETSI Zero-Touch Network and Service Management.
		[186]	HFL	Multi-discriminator bidirectional Wasserstein GAN-gradient penalty algorithm.	1) FL-based three-tier distributed VM anomaly detection framework. 2) Enables distributed network slice managers to collaboratively train a global VM anomaly detection model. 3) Consider the powerful ability of GAN to address the high-dimensional, and imbalanced data.	Complexity of the Multi-discriminator bidirectional Wasserstein GAN-gradient penalty algorithm may make the approach challenging to implement.	-
	FL-based cache management over MEC servers	[187]	CFL	-	1) Employ the FL to predict the UEs' demands for a particular content (i.e., future content popularity). 2) Based on the prediction model, the content placement aiming for improved QoS.	1) Increased computational load on UEs due to the FL process. 2) The challenge of unbalanced and non-IID data across UEs for the FL model's training and convergence.	Consider an SDN-controller as the global aggregation point for the FL.
		[192]	CFL	-	1) Federated DRL-based cache replacement aiming for striking balance between similarity cost, latency cost, and cache hit cost. 2) Integrate the recommender systems (i.e., direct hit and soft hit)	1) The challenge of unbalanced and non-IID data across MEC servers for the FL model's training and convergence. 2) Actor-critic DRL model can be sensitive to the choice of hyperparameters.	-

3) Networking Resource Management: FL-based approaches have been extensively studied in the field of networking resource management, particularly for tasks such as bandwidth allocation, power control, and handover management. Given the sequential nature of decision-making in networking resource management, a significant focus of research has been on applying DRL in this field, known as federated DRL. These methods enable collaboration among multiple agents to create a global policy model, effectively addressing privacy and scalability issues, particularly in scenarios with multiple service providers. While most studies primarily utilized CFL, some have explored VFL for different service types, such as network slices. As FL management in this field, FL participant selection and the use of Shapley values for aggregation are employed to minimize communication overhead in the FL process. Nevertheless, challenges with unbalanced and non-IID data among FL participants still exist, and the field lacks comprehensive FL management solutions to address the challenges compared to the mobile traffic prediction domain. Only a few studies have addressed these issues, proposing FL mechanisms such as incorporating DQN parameters and a Proxy-Lagrangian framework. Specifically, in FL-based handover management, UEs are considered as FL participants for global policy model generation. The limited computing and battery capacities of UEs pose a significant challenge to the FL process, necessitating careful strategies to either reduce this burden or provide incentives to the UEs.

4) Computing Resource Management: FL-based computing resource management has emerged as a prominent area of research, encompassing tasks such as computing resource scaling/provisioning, network slice placement, network slice orchestration, VM anomaly detection, and cache management. Similar to networking resource management, many studies have employed federated DRL for effective sequential decision-making, while some still rely on predictive models for future demand to guide heuristic-based resource management. However, as the application of FL in these areas is still in its infancy, considerations for FL management and diverse FL methods remain limited. Exploratory work is being done on DFL approaches to facilitate decentralized FL processes without a centralized parameter server. Nonetheless, given DFL's sensitivity to aggregation methods, further investigation into convergence issues with varying aggregation methods is necessary. The adoption of a proxy-Lagrangian framework is seen for constraint satisfaction during FL training, but it adds complexity and requires realistic testbed implementation to prove its viability. In FL-based VM anomaly detection, a GAN approach is employed to handle high-dimensional and imbalanced data, though this introduces additional complexity and calls for practical discussion. Cache management via FL also considers UEs with limited computing and battery power as FL participants, driving a need for strategies to reduce their burden or provide incentives. Overall, similar to other domains, FL in computing resource management faces inherent challenges in convergence and computing

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TABLE X
SUMMARY OF RESEARCH CHALLENGES AND POSSIBLE DIRECTIONS

Challenges	Description	Possible directions
Practical deployment issues	To make FL a practical reality, we need to address standardization perspectives and enhance the comprehensibility of FL models. Research efforts should focus on developing standardized practices and making FL models more transparent and interpretable.	• Design a detailed procedure and interface to support interoperability among multiple NWDAFs for FL. • Consider vertical FL from various types of data from different NFs for mobile network management • Explore a range of Explainable AI (XAI) techniques to enhance explainability, transparency, and trustworthiness when deploying FL for mobile network management in a practical manner.
Scalability issues	FL require a significant computational burden on FL participants, particularly NFs on MEC servers with limited computing power. Investigating scalable training mechanisms is essential to ensure FL's effectiveness.	• Hybrid federated and centralized learning (HFCL) can be applied as novel architectures for mobile network management by utilizing both centralized learning and FL depending on network entities' computing capability and data privacy level. • Explore the DFL as a means to achieve full decentralization without relying on a centralized server for mobile network management. Since improving convergence speed and accuracy in the DFL is particularly challenging, it involves the design of an efficient DFL framework for mobile network management. • Examine the potential application of split federated learning (SFL) in the context of mobile network management. Investigate various methods for selecting cut layers and implementing differential privacy within SFL to optimize performance factors such as privacy, latency, and energy efficiency in the context of mobile network management.
Privacy and security issues	Privacy concerns have been raised due to potential data leakage from FL models, particularly in the presence of multiple MVNOs sharing infrastructure. Additionally, the threat of malicious network entities injecting poisoned or noisy models must be mitigated.	• Robust approaches like differential privacy can be employed in mobile network management. Given its inherent computational complexity, it is essential to carefully balance the trade-off between ensuring privacy and maintaining system performance. • Anti-jamming schemes such as frequency hopping, which entail transmitting multiple copies of model updates across different frequencies, can be implemented. Furthermore, the integration of blockchain technology is a viable option. In this regard, protection mechanisms for mobile network management should be devised with a focus on achieving a balanced trade-off between model accuracy and system performance.
OPEX issues	In 6G, due to its denser networks, controlling OPEX, including energy expenditure, becomes more critical. FL should be managed in a cost-effective manner to ensure its long-term viability.	• Energy-efficient FL for mobile network management requires the exploration of sophisticated FL management techniques, including dynamic control of local iterations, FL participant selection, and more. • Spiking neural networks (SNNs), which are emerging as alternatives to deep neural networks, offer the potential for energy-efficient model training within the FL framework, as opposed to traditional deep learning. • There is a need for research into accuracy loss correction and model compression in the context of FL, with the aim of enhancing cost-efficient mobile network management.
Federated reinforcement learning (FRL)	The integration of FL with RL can enable collaborative sequential decision-making in mobile network management, offering new avenues for optimizing network performance.	• RL is commonly used in mobile network management for tasks such as handover and cache management, but practical implementation faces challenges due to large action and state spaces, leading to the need for FRL to enable secure agent model exchange and sample efficiency. • FRL is categorized into horizontal FRL and vertical FRL. Research into vertical FRL for mobile network management, focusing on specific feature integration, is currently limited and warrants further investigation.

load, necessitating more sophisticated FL management approaches.

5) Architecture or Implementation Activities: The prototype of AI-driven Kubernetes orchestration in MEC-enabled mobile networks has been explored to support the FL process. It specifically used WebSocket connections or Apache Kafka for real-time data pipelines. Implementation details of a Docker compose-based FL framework, compatible with container orchestration engines specified in the ETSI NFV IFA 029, were provided to emulate the virtualization environment common in cloud-native deployments. An alternative architecture, DFL, has gained attention for securely deploying the FL framework in 5G networks and beyond. This approach leverages blockchain technology to address the vulnerabilities of centralized model aggregation to server attacks and single points of failure. Blockchain's immutable data storage and decentralized ledger enhance security and trust. For instance, in UAV orchestration for mobile network management, blockchain technology ensures data privacy and trusted exchange. The blockchain-based FL architecture, Skunk's FL

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system, offers benefits like traceability, decentralization, and enhanced security, particularly in detecting attacks in IoT networks within 5G-enabled network-sliced environments.

VI. Research Challenges and Directions

While FL holds significant progress for distributed training in 5G and beyond networks, several key challenges must be addressed to ensure its effective deployment and operation in the context of mobile network management. We consider these challenges across five distinct dimensions such as practical deployment, scalability, privacy and security, OPEX, and FRL issues as presented in Table X. In summary, while FL has tremendous potential, it is essential to address these challenges comprehensively to fully realize its benefits in the context of mobile network management for 5G and beyond networks.

A. Practical Deployment Issues

1) NWDAF for FL-Empowered Mobile Network Management: In the 5G core, the 3GPP standardization body specifies the interfaces and procedures to support NWDAF for AI-empowered mobile network management, as exemplified by subscription requests and analytics information response. Nevertheless, a detailed procedure is yet to be specified to support interoperability among multiple NWDAFs for FL. Furthermore, for implementation concerns, to efficiently manage various NWDAFs for FL, the interface for acquiring information regarding the physical location of NWDAFs and computation/storage capacity should be carefully defined. Moreover, as discussed in [1], most conventional studies cover horizontal FL, in which different sample datasets with the same feature space are used for local training. However, because the NWDAF collects various types of data from different NFs for a specified user (i.e., the same sample datasets with varying features), vertical FL can be widely applied to the 5G core. For example, the first dataset can include registration and authentication information, whereas the second dataset can contain information related to each user's choice of network slice. Nevertheless, most studies need to address the open problem of vertical FL in AI-empowered mobile network management, which could be an important research direction. In another research direction, the 5G core must effectively support AI services between user equipment and application servers. In this direction, an investigation of 5G system support for AI/ML-based Services (FS AIMLsys) was undertaken in 3GPP Release 18 [201], [202]. This study aimed to leverage a 5G system as a platform by providing intelligent transmission support for AI/ML applications [202]. In this context, [75] proposed FL participant selection utilizing NWDAF support, where trajectory prediction is considered.

2) Explainable AI for FL-Empowered Mobile Network Management: Due to the lack of explainability, transparency, and trust inherited from black-box AI models, service providers are concerned with deploying AI-empowered mobile network management systems [203]. For instance, service providers may use AI models to make complex resource management decisions with financial and service-quality implications. Such critical issues open up a new AI approach that is required to make understandable and explainable decisions. In this regard, various XAI techniques have been studied to examine the decisions made by AI black box models by calculating the attributes of every participating feature. In particular, XAI techniques have gained popularity in certain domains, such as driving, image recognition, and speech analysis [204]. Nevertheless, there is a need for further studies in the mobile communication domain that consider XAI techniques that provide explainability in their proposed solutions. The work of [205] first outlined the necessity for explainability in forthcoming 6G networks and then summarized the envisioned XAI for handover and resource allocation problems. The study of [206] provides a comparative analysis of various XAI methods, such as XGBoost and SHapley Additive eXplanation (SHAP), to identify the root cause of SLA violations. The authors of [207] illustrated a vast range of XAI metrics for qualitative assessments to ensure trustworthiness. Therefore, further research must be conducted on XAI-enabled mobile network management by considering distributed and isolated network slicing in 5G and 6G networks. Particularly, because the operator/slice tenant must understand the FL model's behavior at specific times (e.g., busy hours) to trust it for practical deployment, XAI-empowered FL for mobile network management can be a crucial research direction for efficiently handling dispersed network slices and helping any business/organization provide secure operation and trust when deploying AI models [206], [208].

B. Scalability Issues

1) Hybrid Federated and Centralized Learning for AI-Empowered Mobile Network Management: As already discussed in the extensive survey, the FL framework has been widely adopted and has opened up prospects of application for AI-empowered mobile network management by providing benefits for overcoming the considerable overhead of transmitting distributed local datasets to a single centralized node for network management. However, FL requires significant computational resources and sufficient datasets from distributed network entities (e.g., mobile devices, IoT devices, and virtualized NFs over distributed MEC servers). In practice, not all network entities have sufficient computational resources or datasets for local training to participate in the FL process. The heterogeneity of such resources and datasets may result in poor convergence speed and degradation of model accuracy. Recently, as an alternative method to address these issues, HFCL approaches have been studied as novel architectures for FL frameworks that are yet to be applied to network management [9], [200], [209]. Consequently, under the assumption of relieving privacy concerns according to the scenarios, only network nodes with sufficient computational resources participate in the FL process, whereas the remaining nodes send their local datasets to the centralized node for centralized learning. As shown in Fig. 16, HFCL utilizes both centralized learning and FL by designating clients as active or inactive, depending on their computational capability. The work of [209] is considered a client for conducting centralized learning and FL simultaneously. Therefore, the

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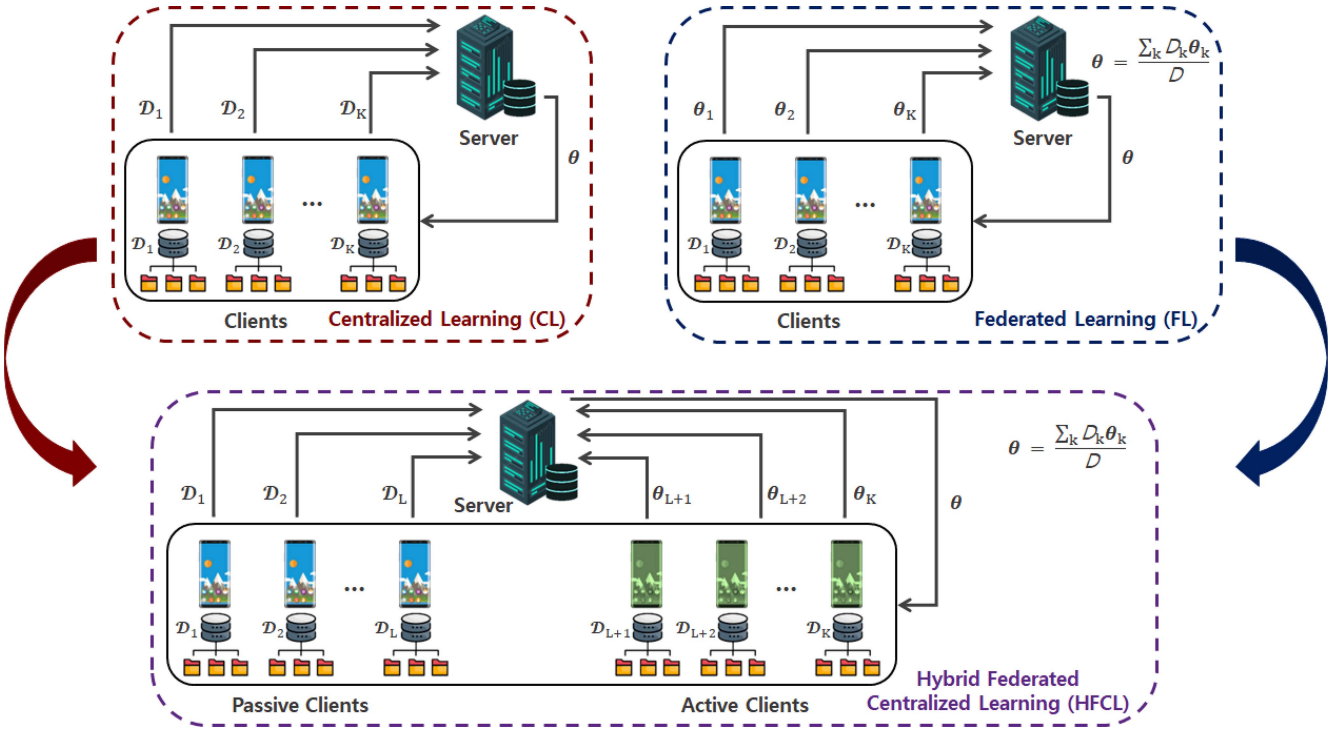


Fig. 16. CL, FL, and HFCL frameworks [200].

authors suggested an optimal data size for centralized learning within the client to minimize the overall learning delay in achieving the target accuracy.

2) Decentralized Federated Learning for Mobile Network Management: DFL, such as gossip learning, represents a fully decentralized learning paradigm that eliminates the need for a centralized parameter server, a requirement in conventional FL. The evident advantages of DFL lie in its infrastructure-free nature and the absence of a single point of failure [210]. By adopting such decentralized frameworks, mobile network management can achieve full decentralization, resulting in cost-effective scalability and improved robustness. For example, in a UAV-assisted MEC scenario [35], deploying the FL framework without relying on a centralized MEC server on the ground is possible. Moreover, even distributed MEC servers on the ground, located far from a centralized parameter server, can collaboratively generate a global model in a P2P manner using DFL. Despite the numerous benefits offered by DFL, achieving higher accuracy and faster convergence compared to CFL is more challenging due to the phenomenon of client drift. Researchers have explored various solutions to enhance convergence speed and accuracy in DFL, including aggregation management, clustering, knowledge distillation, mutual knowledge transfer, and others [211], [212]. However, designing an efficient DFL framework for mobile network management remains an open and critical research direction that demands further attention. The distributed and heterogeneous nature of network entities adds complexity to this problem, making it both challenging and promising for future investigations.

3) SFL for Mobile Network Management: To alleviate the computational burden on participants in FL, an innovative technique known as split learning (SL) has emerged as a practical solution. SL involves dividing a full deep learning model into two sub-models, which can be trained both at a central server and across multiple distributed SL participants. While SL offers substantial advantages, its sequential training approach can lead to inefficient distributed processing and slow convergence. In response to these challenges, a novel approach called split federated learning, or simply SFL, has recently been introduced. SFL seeks to leverage the strengths of both FL and SL by introducing an additional federated learning server and allowing SFL participants to train their local models in parallel. As shown in Fig. 17, the SFL framework includes two key servers: i) a federated learning server (referred to as the fed server) and ii) a main server. In addition, there are multiple clients, known as SFL participants. The fundamental structure of this framework involves the division of the full model into two distinct sub-models such as i) the client-side model, and ii) the server-side model. At each global epoch, each SFL participant performs forward propagation on its local dataset using the client-side model. Subsequently, the SFL participants pass the smashed data and corresponding labels to the main server. The main server then undertakes forward propagation using the server-side model, followed by back-propagation from computing the loss, all of which can be executed in parallel. Following this step, the server-side model is updated utilizing the specific aggregation algorithm, and the main server transmits the gradient of the smashed data back to each client for updating their respective client-side local models. Subsequently, the fed server receives the updated client-side local models from all clients and performs an aggregation operation to generate the client-side global model. Finally, the fed server transmits

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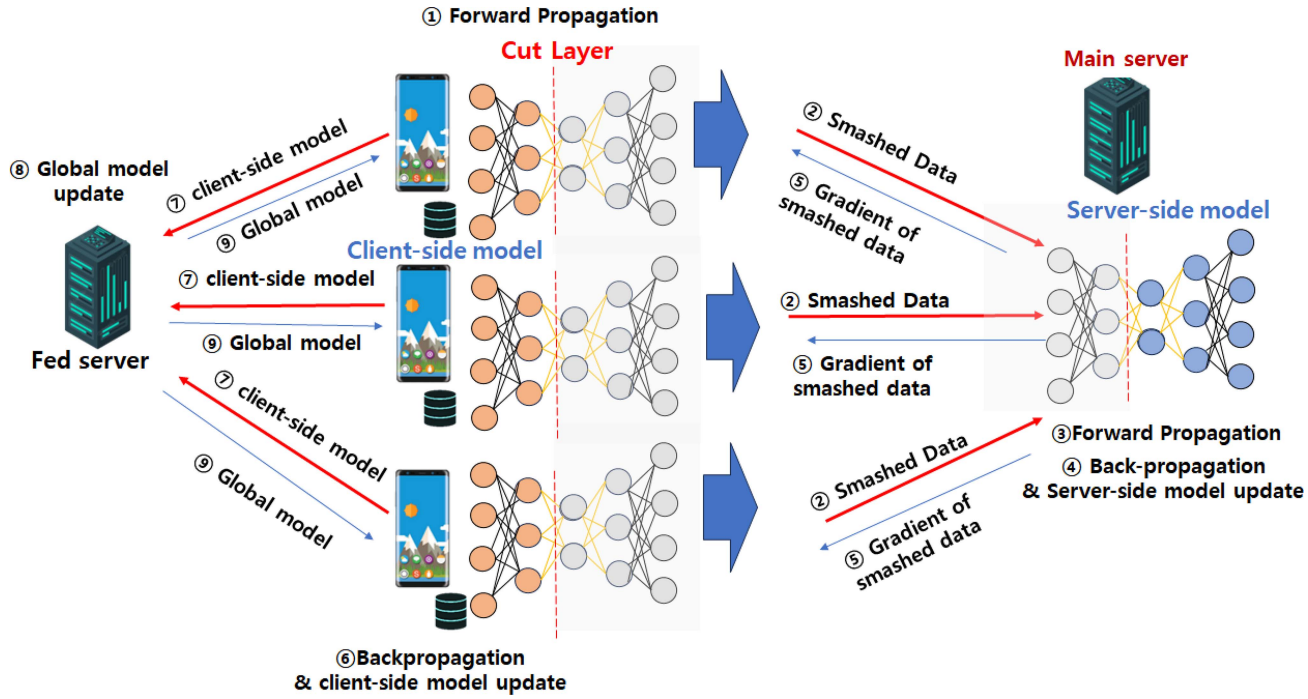


Fig. 17. Workflow of SFL framework.

the updated client-side global model to all clients, facilitating the synchronization of client-side models across the entire SFL framework. This innovative approach shows significant promise for various applications [213], [214]. SFL can offer efficient methods for training deep learning models on NFs with limited computational power. This can be achieved by alleviating the computational burden on NFs while simultaneously preserving their privacy. Correspondingly, these mechanisms can be applied to various aspects of mobile network management, encompassing mobile traffic and mobility prediction, SFL-based modulation recognition applications, and the efficient allocation and management of radio and computing resources in both core and access networks. One particularly promising application of SFL in mobile network management is its ability to support distributed training in scenarios involving MEC-assisted UAVs. In such cases, SFL plays a critical role in reducing the training workload on local UAVs, which often face constraints related to battery capacity and computing power. Moreover, for efficient SFL management, various methods for selecting cut layers and implementing differential privacy within SFL can be investigated to optimize performance factors such as privacy, latency, and energy efficiency in the context of mobile network management.

C. Privacy and Security Issues

1) Privacy and Confidentiality Preservation: FL is widely recognized for safeguarding the privacy of each FL participant by conducting local model training without exposing raw data and exchanging only updated model parameters. Nevertheless, recent concerns have emerged regarding the potential for local models to disclose sensitive information during communications, leaving them vulnerable to adversaries or third parties who may try to extract raw data [6]. This concern becomes particularly critical in scenarios like mobile network management, where shared infrastructures host multiple MVNOs. Due to the increased adoption of high-frequency networks like 5G and beyond, multiple MVNOs often deploy and share infrastructures to overcome limited BS coverage. For efficient resource allocation and network management policy, these MVNOs can collaborate through the FL process. To ensure privacy and security, robust security solutions such as differential privacy and collaborative training should be applied in FL-empowered mobile network management [6]. This challenge is amplified in UAV-aided MEC systems, where model updates from UAVs are transmitted via wireless channels. Nie et al. [35] proposed the use of Gaussian differential encryption to protect the uploading messages during local model exchanges from UAVs. However, adopting such robust security measures sacrifices model accuracy and requires significant computation from FL participants. It is crucial to strike a well-balanced tradeoff between privacy guarantee and system performance while implementing FL-empowered mobile network management, especially considering the limited computation resources available to FL participants who are MEC servers or BSs serving other service requests from NFs or UEs.

2) Poisonous or Noisy Model Issues: Malicious BSs or MEC servers pose a threat by potentially injecting poisoned or noisy models into an FL server. Such instances may occur if some of the BSs or MEC servers are infected with viruses while participating in the FL process. Additionally, model poisoning can also arise from distributed Denial-of-Service (DoS) and jamming attacks conducted by attackers aiming to degrade mobile network management performance [6]. The vulnerability is particularly pronounced during local model updates via wireless channels, such as in UAV-assisted MEC

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server scenarios, due to the exposed nature of the wireless medium, which facilitates easier model poisoning compared to wired mediums. To mitigate this issue, anti-jamming schemes like frequency hopping, involving the transmission of multiple copies of model updates over different frequencies, can be adopted. Moreover, blockchain technology can be leveraged, as demonstrated in [215], to audit all local updates during the FL process, thereby preventing malicious FL participants. The integrity of records on the blockchain ensures that any malicious activities of FL participants can be traced back, as the records are untampered. In this context, the protection mechanisms for FL-empowered mobile network mechanisms should be designed aiming for a balanced tradeoff between model accuracy and system performance.

D. OPEX Issues

1) Energy-Efficient FL-Empowered Mobile Network Management: Energy efficiency is a critical metric for cost-effective AI-empowered mobile network management, which reduces the OPEX of service providers. While conventional optimization algorithms, such as the matching theory or other heuristics, require high computational complexity and result in latency, the computational complexity of data-driven AI algorithms is transferred from online optimization to offline training. Therefore, the energy consumption on the network side rapidly increases as the size of the AI model increases (i.e., as the number of variables to be predicted increases). Thus, an energy-efficient FL framework can be designed to cost-effectively automate mobile network management using AI technologies by considering system-wide issues to improve energy efficiency. During the FL process, the overall energy consumption involves three main components: i) Computational energy consumption by FL participants (e.g., mobile devices, MEC servers) during local training. ii) Transmission energy consumption for delivering local model and global model updates. iii) Computational energy consumption for model aggregation. In this context, energy consumption for model aggregation is relatively small. To effectively reduce computational energy consumption during local training and transmission energy, sophisticated FL management techniques should be designed. These may include dynamic control of local iterations, FL participant selection, and more, aiming to decrease the OPEX of service providers. For example, increasing the number of local iterations may result in a smaller number of required rounds for convergence. However, this comes at the expense of higher energy consumption and latency per round. Balancing the transmission energy and computational energy is essential for minimizing the total energy consumption through sophisticated management strategies. By finding the right balance, energy-efficient FL processes can be achieved. However, most conventional studies on energy-efficient FL frameworks have focused on reducing the power consumption of mobile devices with limited batteries. There is a lack of research on energy-efficient FL frameworks for mobile network management on a system-wide scale. This gap exists because most studies focus on energy consumption issues in scenarios with limited battery power, such as mobile devices. However, it is important to recognize that energy efficiency concerns in mobile network management remain significant for reducing OPEX when achieving network management automation. Furthermore, third-generation neural network models can also be utilized to improve the energy efficiency in the training phase, rather than using conventional deep neural networks. Recently, spiking neural networks (SNNs), which use discrete spikes to transmit information and are similar to the information-processing mechanisms of the human brain, have been suggested as energy-efficient alternatives to deep neural networks [216], [217], [218]. From an application perspective, as the FL process involves multiple distributed nodes (e.g., NFs, mobile devices, and BSs) for AI-empowered mobile network management, there is vast scope for leveraging the energy efficiency provided by SNNs. Correspondingly, [30], [219] applied an SNN to the FL framework. In the first study on this issue, the authors of [219] designed an FL framework for the decentralized and privacy-preserving training of SNNs. Furthermore, [30] proposed a decentralized energy-efficient computing method based on SNN that enables edge devices to exploit brain-like biophysiological structures to collaboratively train a global model in the FL process. They revealed that adopting SNN reduces energy consumption by more than 4.5 times compared with the standard FL process. Despite its importance as an early investigation stage for SNN-based FL frameworks, training SNNs in FL-empowered mobile network management has not been researched extensively.

2) Accuracy Loss Correction and Model Compression: Because of the distributed and heterogeneous NFs connected via wired and wireless links, it is essential to train FL models over NFs cost-effectively by considering accuracy loss correction and model compression methods that can be applied to wired and wireless connected NFs or MDs. In this regard, some studies on wireless transmission environments have investigated the theory of accuracy loss correction and model compression (i.e., network model compression and parameter compression) [220]. Thus, various accuracy loss corrections and model compressions must be further investigated for the FL framework over NFs by considering specific reliability constraints that depend on the mobile network management scenario.

E. Federated Reinforcement Learning for Mobile Network Management

RL has been widely adopted in sequential mobile network management decision-making problems such as handover management, cache management, radio resource management, and computation resource scaling. However, implementing RL in practical scenarios poses several challenges. For example, when the action and state spaces are large, the performance of agents is affected by the collected samples. This is because it is nearly impossible to explore all sampling space. Therefore, promoting information exchange among multiple agents while protecting agent privacy is necessary for implementing RL in the real world. These challenges have led to FRL, in which FL

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and RL are integrated under a secure agent model exchange. FRL can be classified into horizontal and vertical FRL, depending on the distribution characteristics of the data, similar to the data partitioning for FL [221]. The horizontal FRL can be applied in scenarios where agents may be distributed geographically; however, they face similar decision-making tasks and have very little interaction with each other in observed environments, such as autonomous driving decision-making model generation. In a horizontal FRL, agents share their experiences by exchanging masked models to enhance sample efficiency and accelerate the learning process. In this regard, [222] proposed FRL-based computation offloading management for IoT systems, where IoT devices train the RL model, and the updated RL models can be aggregated at the edge server by utilizing the FL framework. In contrast, in the vertical FRL, samples of multiple data sets have different feature spaces, but these samples may belong to the same groups or common users. The role of FL in the vertical FRL is to aggregate the partial features observed by various agents. Currently, there are limited studies on vertical FRL compared with horizontal FRL. Therefore, in addition to the horizontal FRL, vertical FRL for mobile network management needs to be further investigated, considering the specific feature integration, such as state/observation and action spaces, of the agents in the mobile network management scenario.

VII. Conclusion

This article has presented a tutorial on FL and a comprehensive study of the challenges involved in FL-empowered mobile network management. This work was motivated by the need for a comprehensive survey on the use of FL for 5G and beyond mobile network management. To bridge this gap after reviewing the state of the art in FL, we have explained the rationale for AI-empowered mobile network management, and how FL can enable cooperative model training for mobile network management at both access and core networks, including distributed NWDAF, mobile traffic or other predictions, and core/access network management, in terms of standardization and research activities. Several main lessons learned from this extensive survey have been summarized and analyzed. Lastly, the paper has explored future research areas and challenges.

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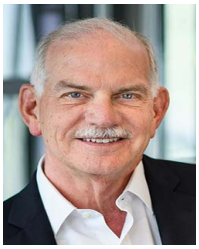
Joohyung Lee (Senior Member, IEEE) received the B.S., M.S., and Ph.D. degrees from the Korea Advanced Institute of Science and Technology, Daejeon, South Korea, in 2008, 2010, and 2014, respectively. He is currently an Associate Professor with the School of Computing, Gachon University and also a Visiting Fellow with the Department of Electrical and Computer Engineering, Princeton University. From 2012 to 2013, he was a Visiting Researcher with the Information Engineering Group, Department of Electronic Engineering, City University of Hong Kong, Hong Kong. From 2014 to 2017, he worked as a Senior Engineer with Samsung Electronics. He has contributed several articles to the International Telecommunication Union Telecommunication and 3rd Generation Partnership Project. His current research interests include federated learning, 6G networks, cloud/edge computing, and network economics.



Faranaksadat Solat received the B.S. degree in industrial engineering from Alzahra University and the M.S. degree in information technology engineering from the Iran University of Science and Technology, Tehran, Iran, in 2017 and 2019, respectively. She is currently pursuing the Ph.D. degree with the School of Computing, Gachon University, South Korea. Her research interests include system optimization, machine learning, federated learning, 5G/6G network management, and reinforcement learning.



Tae Yeon Kim received the B.S. and M.S. degrees in computer engineering from Chung-Ang University, Seoul, Republic of Korea, in 1990 and 1992, respectively, and the Ph.D. degree in network engineering from Chungbuk National University, Cheongju, Republic of Korea, in 2008. He currently leads the Network Intelligence Research Section, ETRI, Republic of Korea. Since 2012, he has worked in programmable networks with network function virtualization and software-defined networking technologies. His research interests include AI-native networking technologies for 6G networks.



H. Vincent Poor (Life Fellow, IEEE) received the Ph.D. degree in EECS from Princeton University in 1977. From 1977 to 1990, he was on the faculty of the University of Illinois at Urbana–Champaign. Since 1990, he has been on the faculty at Princeton, where he is currently the Michael Henry Strater University Professor. From 2006 to 2016, he served as the Dean of Princeton's School of Engineering and Applied Science. He has also held visiting appointments at several other universities, including most recently at Berkeley and Cambridge. His research interests are in the areas of information theory, machine learning and network science, and their applications in wireless networks, energy systems, and related fields. Among his publications in these areas is the recent book *Machine Learning and Wireless Communications* (Cambridge University Press, 2022). He is a member of the National Academy of Engineering and the National Academy of Sciences and is a foreign member of the Chinese Academy of Sciences, the Royal Society, and other national and international academies. He received the IEEE Alexander Graham Bell Medal in 2017.

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